LLM ENGINEERING

Week 03

Multi-Head Attention & Positional Encoding

Thursday, 29 October 2026

Interactive version: https://llm.strueber.dev/#/week-03

Bethgelab · University of Tübingen · Winter term 2026/27

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Recap & Roadmap

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Announcements

Next week, 5 November

- Lecture in person again
- Read Bonus 2 before the tutorial

Quiz 1 on ILIAS

- Weeks 1, 2 and Bonus 1
- Due Friday 30 October, end of day

Second exercise: this lecture

- Due Wednesday 4 November, 22:00
- You get an email when it arrives

▼ Dates for the coming week

Next week's lecture, on 5 November, takes place in person again.

Quiz 1 is on ILIAS and closes at the end of Friday, 30 October. It covers Week 1, Week 2 and Bonus 1.

The second coding exercise covers the content of this lecture and is due on Wednesday, 4 November, at 22:00. You will get an email when it is in your pair's repository.

Bonus 2, the chapter on long contexts, is required reading. The Week 4 tutorial discusses its reflection questions, so read it before then.

Week 2 built one head and one block

- One attention head, width `D`
- One pre-norm block, final RMSNorm, tied head
- Position added once, at the input
- Each query: one softmax row

▼ **Week 2's model gives each query one softmax row**

Last week you built a complete language model around a single attention head. Write B for the batch size, S for the number of prediction positions, D for the model width and V for the vocabulary size. A batch began as raw token windows of shape `[B,S+1]`. One shift turned each window into inputs and next-token targets, both of shape `[B,S]`. A token lookup gave each input token a vector of width D, and a sinusoidal position signal was added to it, so position entered the model once, at the input. The result was the hidden stream X of shape `[B,S,D]`. One pre-norm block transformed the stream without changing its shape. A final RMSNorm and the output head, which is tied to the embedding table, then produced logits of shape `[B,S,V]`.

The block held one attention head of width D. You also estimated what the model costs in parameters, FLOPs and memory, and checked each estimate against a measurement. This week keeps that habit.

For each query, that single head computes one row of softmax weights over the keys up to the query's position, and its output is one weighted average of the value vectors.

This week extends three Week 2 choices: the number of heads, the number of blocks and the way position enters.

How does attention read several things at once, and know where they are?

- Heads: several reads per query
- Shared key and value heads make the cache smaller
- One mask for padded, packed batches
- Position inside the score: rotary embeddings
- Exercise: train six variants and compare them

▼ Today's question

The question for this lecture is: how does attention read several things at once, and know where they are? The Week 2 head gives each query one weighted average. It learns about position only from a signal added at the input. This lecture changes both, in six parts.

1. Attention gets several heads, so that each query gets several reads at the width of one.
2. Query heads share key and value heads, which makes the cache of earlier tokens' keys and values smaller during generation.
3. One composed mask makes attention correct for real batches, which contain padded sequences and several documents packed into one row.
4. The Week 2 block is repeated into a stack of layers.
5. Rotary position embeddings (RoPE) put position into the attention score itself, instead of adding it once at the input.
6. The last part asks what a causal model can learn about position when it receives no explicit position signal.

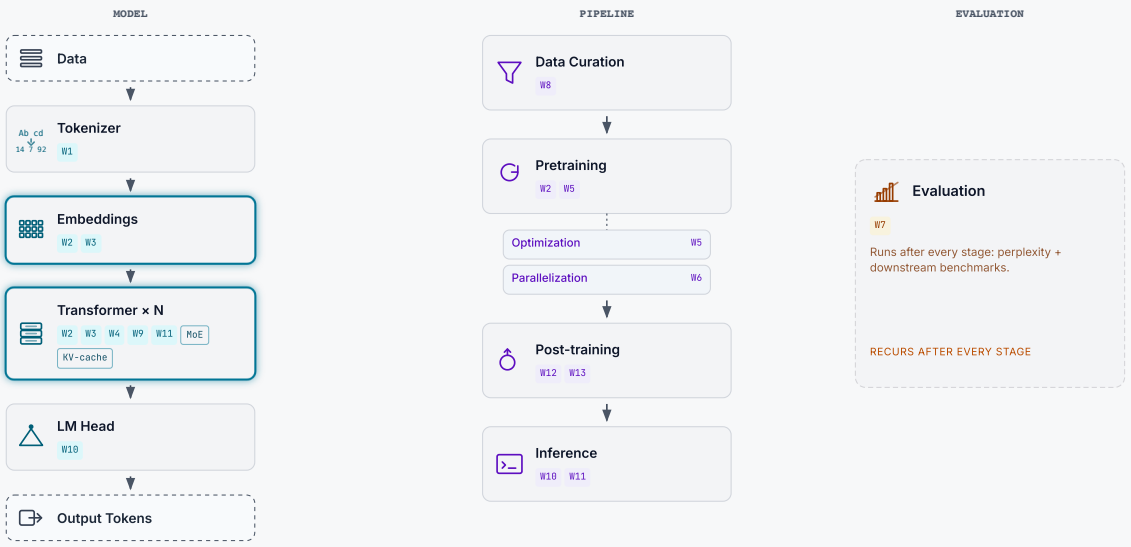
Heads, shared key and value heads, the mask and RoPE are treated in depth. The stack and the model without position are short.

The exercise builds all six parts and trains six variants of the same model: the base model with four heads and RoPE, a single-head model, two models whose four query heads share two key and value heads or only one, and two models with sinusoidal position or none. Every variant trains for the same steps on the same data in the same order, so the variants differ only in their shape. You then compare what each variant costs against what it learns.

Where this fits

Read on your own

- Today: attention, position, the stack
- Week 4: which keys a query reads, and how fast



10 CONNECTIONS

◀ Prev | Reset | Start ▶

Click "Start" to walk through how decisions in one part of the pipeline shape every other part.

▼ Reading the Week 3 pipeline map

The map places this week in the course. Its highlighted stages are the two model stages that change today: the embeddings and the Transformer × N.

Inside the Transformer, multi-head and grouped-query attention reorganize the attention channels. The composed mask changes which query-key pairs are allowed. The repeated block turns the `× N` label into an actual stack.

Position handling touches both stages. Sinusoidal position belongs to the embeddings, rotary embeddings act inside attention, and a model without position uses neither.

The map shows how the course is organized and which parts depend on which. It is a navigation aid and contains no measurements of speed, quality or cause.

Today's changes are attention, position and the stack. Week 4 stays inside attention and asks which keys a query may read and how fast attention runs.

Multi-Head Attention

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How can attention read two places at once without blending them?

Answer A1

The Idea: Attend in Parallel Subspaces

```
Q, K, V     [B,S,D]
reshape     [B,S,H,D_h]
transpose   [B,H,S,D_h]
scores      [B,H,S,S]
context     [B,H,S,D_h]
transpose   [B,S,H,D_h]
concat      [B,S,D]  → W_O
```

- **Multi-head attention** (MHA): head width `D_h = D/H` (Vaswani et al., 2017)
- Each head gets a slice of the learned projections
- Scale by `√D_h`, not `√D`

▼ The shape path and the scale of each head

Let X, of shape `[B,S,D]`, be the input to attention, with batch size B, sequence length S and model width D. Let H be the number of heads and D_h the number of channels per head, which the exercise code calls `head_dim`. The course sets $D = HD_h$, so $D_h = D/H$. That equality is a choice of the course configuration, and other attention designs need not make it.

The course uses three bias-free projections $W_Q, W_K, W_V \in \mathbb{R}^{D\times D}$, each mapping X to a tensor of shape `[B,S,D]`. Multi-head attention (MHA) splits the output of each projection into H heads and runs attention in every head in parallel (Vaswani et al., 2017). Head h, for $h = 0, \ldots, H-1$, takes the channels hD_h to $(h+1)D_h - 1$. Its queries are $Q_h = XW_Q^{(h)}$, where $W_Q^{(h)} \in \mathbb{R}^{D\times D_h}$ is the matching block of columns of W_Q, and its keys K_h and values V_h are formed in the same way. For one sequence, one MHA layer computes

$$C_h = \operatorname{softmax}\left(\frac{Q_h K_h^\top}{\sqrt{D_h}} + M\right) V_h, \quad h = 0, \ldots, H-1,$$
$$O = \begin{bmatrix} C_0 & C_1 & \cdots & C_{H-1} \end{bmatrix} W_O,$$

where M is the causal mask of Week 2, the softmax runs across the keys of each row, the brackets concatenate along the channels, and $W_O \in \mathbb{R}^{D\times D}$. Week 2 called the blended values H. Now that H counts heads, the blended values of head h are its **context** C_h, and the concatenation is the full context C. Week 2's d_k is D_h with $H = 1$: its single head was as wide as the model, $d_k = D$.

An implementation may fuse the three projections into one matrix $W_{QKV} \in \mathbb{R}^{D\times 3D}$, whose output is split into Q, K and V. It holds the same $3D^2$ parameters as three square matrices and computes the same result.

The shape path

The slide's path computes the same definition, batched over the B sequences and the H heads. The reshape to `[B,S,H,D_h]` splits the D projected channels of each token into H groups of D_h, so each head receives a learned slice of the projected channels and sees all of X through its own weights. The transpose to `[B,H,S,D_h]` moves the head axis in front of the sequence axis. Each head is then an independent attention computation over S positions. The query-key products give the scores `[B,H,S,S]`, one $S \times S$ matrix per head, and multiplying the softmax weights by V gives the context `[B,H,S,D_h]`.

The way back takes two steps. A transpose first returns the context to `[B,S,H,D_h]`, and a reshape then concatenates the heads of each position into `[B,S,D]`. The transpose back is the step a reshape-only version gets wrong. Reshaping `[B,H,S,D_h]` straight to `[B,S,D]` produces the right shape with the wrong content: each output row then holds several positions of one head, instead of all heads of one position. Finally W_O maps the concatenated heads to the block output `[B,S,D]`.

A synthetic example checks the shapes, with $B = 2$, $S = 4$, $D = 8$ and $H = 2$, so $D_h = 4$:

Step	Shape
input X	`[2,4,8]`
Q, K and V, each	`[2,4,8]`
reshape	`[2,4,2,4]`
transpose	`[2,2,4,4]`
scores	`[2,2,4,4]`
context	`[2,2,4,4]`
transpose back	`[2,4,2,4]`
concatenate	`[2,4,8]`
output of W_O	`[2,4,8]`

The scores and the context have the same shape here only because the example has $S = D_h = 4$. Their last axes still mean different things: keys for the scores, channels for the context.

The scale follows the width of a head

Each head divides its scores by $\sqrt{D_h}$, because its dot product runs over D_h coordinates. Week 2 derived the scale for one head of width d_k, and the same argument applies to each head separately.

- **A1. Week 2's variance-preserving assumptions.** Under Week 2's A1 to A3, every coordinate of a head's query and key is a coordinate of the full XW_Q or XW_K, with mean 0 and variance about 1 at initialization.
- **A2. The course split.** The heads together have $HD_h = D$ channels.

Recall Week 2's result for a query and a key of d_k coordinates, under Week 2's A1 to A3:

$$\mathrm{Var}(q \cdot k) \approx d_k. \tag{F1}$$

Recall also how a variance scales under a constant factor c:

$$\mathrm{Var}(cY) = c^2 \,\mathrm{Var}(Y). \tag{F2}$$

Claim. Under A1 and A2, at initialization, the unscaled score $q \cdot k$ of one query and one key of a head has a variance of about D_h. Dividing it by $\sqrt{D_h}$ brings the variance to about 1. Dividing by $\sqrt{D}$ instead makes every score $\sqrt{H}$ times smaller, with a variance of about $1/H$.

Here q and k each have the D_h coordinates of one head:

$$\begin{aligned}
\mathrm{Var}(q \cdot k) &\approx D_h && \text{F1 with } d_k = D_h, \text{ by A1} \\
\mathrm{Var}\left(\frac{q \cdot k}{\sqrt{D_h}}\right) &\approx 1 && \text{F2 with } c = 1/\sqrt{D_h} \\
\frac{q \cdot k}{\sqrt{D}} &= \frac{1}{\sqrt{H}} \cdot \frac{q \cdot k}{\sqrt{D_h}} && \text{A2: } \sqrt{D} = \sqrt{H}\,\sqrt{D_h} \\
\mathrm{Var}\left(\frac{q \cdot k}{\sqrt{D}}\right) &\approx \frac{1}{H} && \text{F2 with } c = 1/\sqrt{H}.
\end{aligned}$$

The second line is the order-one scale that Week 2's division restores, and the third line is exact for every score of every head, because it uses only A2. The course's actual initialization gives each coordinate a variance of about $1/3$ instead of 1, so Week 2 found $\mathrm{Var}(q \cdot k) \approx D_h/9$. That scales the first, second and fourth lines by $1/9$, but not the exact third line. In Week 2 the single head had $d_k = D$, and the two divisors coincided. With $H > 1$ they differ. At the course configuration, $D = 512$ and $H = 4$, the divisor is $\sqrt{128} \approx 11.3$, not Week 2's $\sqrt{512} \approx 22.6$. Keeping $\sqrt{D}$ is therefore a bug: it divides every score by an extra factor of $\sqrt{H}$, which is 2 here.

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Which has more parameters?

One head of width 512, or four of width 128?

Answer A2

Each head has QK and OV circuits Read on your own

- QK circuit: decides which keys get attention (Elhage et al., 2021)
- OV circuit: decides what gets written back
- Heads can mix several behaviors
- Heads compose across layers, as induction heads show (Olsson et al., 2022)

▼ Two circuits as a way to analyze a head

Elhage et al. analyze an attention head as two circuits (Elhage et al., 2021). The **QK circuit**, formed by the query and key projections, sets the attention scores. It decides which query-key pairs score high, and so which keys a query attends to. The **OV circuit**, formed by the value and output projections, decides how the attended information is transformed before it is written back into the residual stream.

The circuits as matrices

Split W_O by rows into H blocks $W_O^{(h)} \in \mathbb{R}^{D_h \times D}$, one per head, so that block h multiplies the channels of the context C_h in the concatenation. Block multiplication turns the concatenation into a sum,

$$O = \begin{bmatrix} C_0 & \cdots & C_{H-1} \end{bmatrix} W_O = \sum_{h=0}^{H-1} C_h \, W_O^{(h)},$$

so each head writes its own term into the output. Let x_m be the input row at position m. Before the mask and the softmax, the score of head h between query position m and key position n is

$$\frac{x_m \, W_Q^{(h)} W_K^{(h)\top} \, x_n^\top}{\sqrt{D_h}}.$$

Let A_h be the attention weights of head h, of shape `[S,S]` for one sequence. The term the head writes is

$$C_h \, W_O^{(h)} = A_h \, X \, W_V^{(h)} W_O^{(h)}.$$

Without a position rotation, as in this definition, the QK circuit is the $D \times D$ matrix $W_Q^{(h)} W_K^{(h)\top}$, and the OV circuit is the $D \times D$ matrix $W_V^{(h)} W_O^{(h)}$. The QK circuit enters only the weights A_h, so it chooses where the head reads. The OV circuit acts on whatever those weights select, so it sets what the read contributes.

The circuits describe a head's computation, which need not amount to one job a person can name, because a head can combine several behaviors. The output projection W_O also mixes the contributions of all heads, and later layers can build on what earlier heads wrote.

Induction heads compose two layers

Depth lets heads in different layers compose. One studied example is the induction pattern: when a sequence contains `[A][B]` and later `[A]` again, the model predicts `[B]`. A previous-token head composed with an induction head across two layers produces this pattern (Olsson et al., 2022). In the earlier layer, the previous-token head writes into each position which token came before it. In the later layer, the induction head at the second `[A]` compares its query with those writes and attends to the position whose previous token was `[A]`, which is the position of `[B]`. Its OV circuit then copies `[B]` toward the prediction.

The induction example is not a claim about every head, and the exercise does not ask you to find or build an induction head.

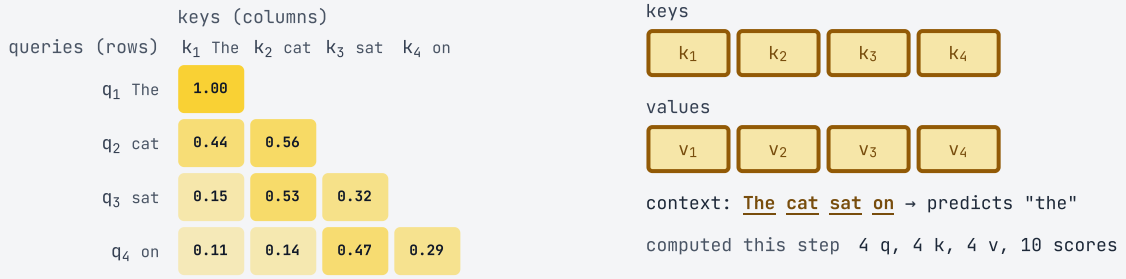
Grouped-Query Attention

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One more token: what is new?

Which scores, keys and values must be computed?

ILLUSTRATIVE | Prefill | Decode 1 | Decode 2



context: The cat sat on → predicts "the"

computed this step 4 q, 4 k, 4 v, 10 scores

computed this step

Rows are queries and columns are keys, on the toy sentence from Week 2's heatmap. A filled cell is computed in the current step, and its number and shade give its attention weight. An empty cell above the diagonal is masked. The two strips list the key and the value of every position the model has seen.

Answer A3

10

Which needs more memory?

Course model, 25M parameters.
Serving 32 sequences of 1,024 tokens in BF16:
the weights or the KV cache?

Answer A4

Every KV head costs cache memory

KV cache

$$\text{KV bytes} = 2BSLH_{kv}D_h \text{ bytes/element}$$

- Cached: K, V of every past token
- Grows with batch, context, depth
- One factor to cut: `H_kv`

▼ The cache scales with every factor of the payload

The slide writes the payload of the previous slide in bytes:

$$\text{KV bytes} = 2\,B\,S\,L\,H_{kv}\,D_h \cdot \text{bytes per element}.$$

Claim. The cache size is proportional to each of its factors: the batch B, the context S, the depth L, the head width D_h, the number of KV heads H_{kv} and the bytes per element of the data type. The formula is a product, so doubling one factor while the others stay fixed doubles the size.

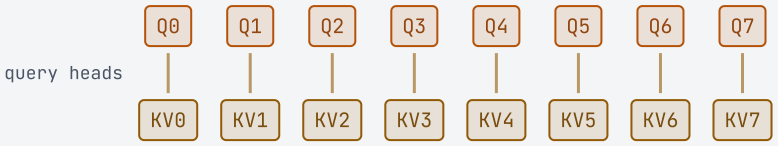
Of these factors, sharing KV heads changes one. It makes H_{kv} smaller, and the cache shrinks in proportion. Whether that saving is worth what the model gives up is a separate question.

The formula counts the cached tensors alone, as an ideal payload, and leaves out the memory traffic of one decode step. A serving system adds paging and allocator metadata, and it may shrink the cache by quantizing it or by keeping only a local attention window. Week 10 covers allocation, eviction and the rest of the cache's lifecycle.

Query heads can share key and value heads

- MHA: `n_kv_heads = n_heads`
- **Grouped-query attention** (GQA): `n_kv_heads` between 1 and `n_heads`, dividing `n_heads` (Ainslie et al., 2023)
- **Multi-query attention** (MQA): `n_kv_heads = 1` (Shazeer, 2019)
- Each KV head serves `g = H/H_kv` query heads

ILLUSTRATIVE variant: MHA GQA MQA



query heads	KV heads	KV cache vs MHA
8	8	8/8 (full)

MHA keeps one KV head per query head. MQA shares a single KV head among all query heads. GQA lies between the two: each KV head serves *g* query heads. Fewer KV heads make the KV cache smaller at inference, which Week 10 covers.

▼ Mapping query heads to shared KV heads

Write `n_heads` as H, the number of query heads, and `n_kv_heads` as H_{kv}, the number of key-value heads. The query projection keeps all H heads, while the key and value projections produce only H_{kv} heads. The projected tensors therefore have the shapes `Q:[B,H,S,D_h]` and `K,V:[B,H_kv,S,D_h]`.

Multi-head attention (MHA) is the case $H_{kv} = H$. Multi-query attention (MQA) is the case $H_{kv} = 1$, in which every query head reads one shared key and value head (Shazeer, 2019). Grouped-query attention (GQA) covers the cases in between, with H_{kv} a divisor of H strictly between 1 and H (Ainslie et al., 2023). Every valid configuration has H divisible by H_{kv}, in code `n_heads % n_kv_heads == 0`, and the course configuration rejects any other.

The group map

Definition. Each KV head serves a group of

$$g = \frac{H}{H_{kv}}$$

query heads, and the groups are consecutive. Query head h, counted from zero, reads KV head

$$h \mapsto \lfloor h/g \rfloor, \qquad h = 0, 1, \ldots, H-1.$$

The exercise requires this consecutive order. For $H = 4$ and $H_{kv} = 2$ the group size is $g = 2$:

Query head h	0	1	2	3
KV head $\lfloor h/2 \rfloor$	0	0	1	1

MHA is the endpoint $g = 1$, where each group holds one query head and query head h reads KV head h. MQA is the endpoint $g = H$, where one group holds all query heads and every one of them reads KV head 0.

Sharing without copying

The query heads of a group reuse its keys and values logically. An implementation may present one KV head to several query heads, for example through a broadcast view, and the cache still stores that head once. If you expand the KV heads to H heads instead, the expansion has to reproduce the group map. The call `repeat_interleave(g, dim=1)` does: it repeats each KV head g times in place, so expanded head h is KV head $\lfloor h/g \rfloor$ and each group stays consecutive. The call `repeat(1, g, 1, 1)` tiles the whole set of KV heads g times, so expanded head h is KV head $h \bmod H_{kv}$. For $H = 4$ and $H_{kv} = 2$:

Expanded head h	0	1	2	3
`repeat_interleave(2, dim=1)`	0	0	1	1
`repeat(1, 2, 1, 1)`	0	1	0	1

With `repeat`, the shapes are right and the pairs are wrong: query head 1 reads KV head 1, which belongs to query heads 2 and 3. The two calls agree at the endpoints. For MHA, $g = 1$ and both leave the heads as they are. For MQA, $H_{kv} = 1$ and both repeat the one head. A test at MHA or MQA therefore cannot catch the wrong order. Only a configuration strictly between them, such as $H_{kv} = 2$, exposes it.

Reading the widget

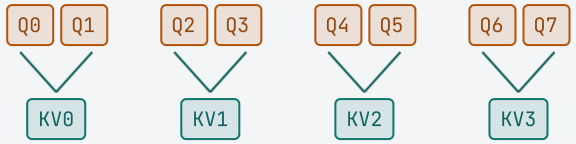
The widget keeps eight query heads and changes only the number of KV heads they share. The last column gives the course model's KV heads in the same state:

State	KV heads, $H = 8$	Readout "KV cache vs MHA"	KV heads, $H = 4$
MHA	8	`8/8 (full)`	4
GQA	4	`4/8 of MHA`	2
MQA	1	`1/8 of MHA`	1

The course model's GQA case, $H_{kv} = 2$, keeps the same fraction of KV heads as the widget's, $2/4 = 4/8 = 1/2$, with the same group size $g = 2$. The fractions compare ideal cache

payloads with the MHA configuration.

ILLUSTRATIVE variant: MHA GQA MQA



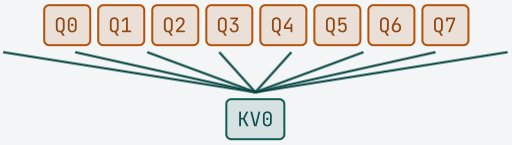
KV heads 4 KV heads

query heads	KV heads	KV cache vs MHA
8	4	4/8 of MHA

MHA keeps one KV head per query head. MQA shares a single KV head among all query heads. GQA lies between the two: each KV head serves *g* query heads. Fewer KV heads make the KV cache smaller at inference, which Week 10 covers.

GQA: each of the 4 KV heads serves 2 consecutive query heads, and the KV cache drops to 4/8 of MHA's.

ILLUSTRATIVE variant: MHA GQA MQA



KV heads 1 KV head (shared)

query heads	KV heads	KV cache vs MHA
8	1	1/8 of MHA

MHA keeps one KV head per query head. MQA shares a single KV head among all query heads. GQA lies between the two: each KV head serves *g* query heads. Fewer KV heads make the KV cache smaller at inference, which Week 10 covers.

MQA: all 8 query heads read one shared KV head, and the KV cache drops to 1/8 of MHA's.

What GQA Saves

- `D=512`, `H=4`, `D_h=128`, `L=8`, `S=1024`, BF16

Variant	projection params / layer	cache `B=1`	cache `B=32`
MHA (`H_kv=4`)	1,048,576	16 MiB	512 MiB
GQA (`H_kv=2`)	786,432	8 MiB	256 MiB
MQA (`H_kv=1`)	655,360	4 MiB	128 MiB

- QK and AV work unchanged
- Speed depends on the implementation
- Quality: compare loss curves (Ainslie et al., 2023)

▼ Reproducing the parameter, cache and work counts

The table holds the course base configuration fixed and changes only the number of KV heads, through $H_{kv} = 4$, 2 and 1.

Assumptions

- **A1. The course base configuration:** width $D = 512$, $H = 4$ query heads of width $D_h = 128$ (so $D = HD_h$), $L = 8$ layers and sequences of $S = 1{,}024$ tokens.
- **A2. Bias-free dense projections:** W_Q and W_O map width D to width D, and W_K and W_V map width D to width $H_{kv}D_h$.
- **A3. BF16 storage:** 2 bytes per cached element, with $1\,\mathrm{MiB} = 2^{20}$ bytes.
- **A4. The MLP width:** two matrices, $D \to D_{ff} \to D$, with $D_{ff} = 3D = 1{,}536$ as in Week 2.

Parameters per layer

Claim. Under A1 and A2, the attention projections of one layer hold

$$P_{\text{attn}}(H_{kv}) = 2D^2 + 2D\,H_{kv}\,D_h \tag{1}$$

parameters:

$$\begin{aligned} P_{\text{attn}}(H_{kv}) &= \underbrace{D \cdot D}_{W_Q} + \underbrace{D \cdot D}_{W_O} + \underbrace{D \cdot H_{kv}D_h}_{W_K} + \underbrace{D \cdot H_{kv}D_h}_{W_V} \quad \text{A2} \\ &= 2D^2 + 2D\,H_{kv}\,D_h. \end{aligned}$$

The Q and O term, $2D^2 = 2 \cdot 512^2 = 524{,}288$, stays fixed, and only the K and V term changes with H_{kv}:

Variant	H_{kv}	Q and O, $2D^2$	K and V, $2D\,H_{kv}\,D_h$	P_{attn}
MHA	4	524,288	524,288	1,048,576
GQA	2	524,288	262,144	786,432
MQA	1	524,288	131,072	655,360

For MHA, $H_{kv}D_h = HD_h = D$, and equation (1) gives back the $4D^2$ of the Multi-Head Attention tab.

Over the whole model, the saving is the change in the K and V term times the $L = 8$ layers. For MQA it is $8 \cdot (524{,}288 - 131{,}072) = 3{,}145{,}728$ parameters, so the model drops from 25,174,528 to 22,028,800 parameters, about 12.5% fewer.

Cache bytes

Claim. Under A1 and A3, at a fixed batch and context the cache is proportional to H_{kv}, and sharing KV heads divides it by the group size $g = H/H_{kv}$.

Compare the payload of "Weights against cache in one serving setup" with its MHA value:

$$\frac{2\,B\,S\,L\,H_{kv}\,D_h}{2\,B\,S\,L\,H\,D_h} = \frac{H_{kv}}{H} = \frac{1}{g}.$$

Every other factor cancels. For one sequence of the MHA model, the payload in bytes is

$$2 \cdot 1 \cdot 1024 \cdot 8 \cdot 4 \cdot 128 \cdot 2 = 16{,}777{,}216,$$

which is 16 MiB. At $B = 32$ each payload is 32 times larger:

Variant	H_{kv}	Bytes, $B = 1$	MiB, $B = 1$	MiB, $B = 32$
MHA	4	16,777,216	16	512
GQA	2	8,388,608	8	256
MQA	1	4,194,304	4	128

The work of one block

Definition. One scalar multiply-accumulate counts as one MAC. The count covers one block's forward pass: the bias-free Q, K, V and O projections, the two $S \times S$ matrix products of attention, and the two MLP products. It takes both attention products in full, even though the causal mask blocks the upper triangle. It leaves out everything else, from embeddings and softmax to the backward pass and kernel effects.

The two attention products are the scores $QK^\top$ and the weighted sum AV, where A holds the attention weights of shape `[B,H,S,S]`. Recall that multiplying an $m \times n$ matrix by an $n \times p$ matrix takes

$$mnp \text{ MACs.} \tag{F1}$$

Claim. Under A1, A2 and A4, the projections, the attention products and the MLP of one block take

$$\begin{aligned}
M_{\text{proj}} &= BS\, P_{\text{attn}}(H_{kv}) && \text{F1} \\
&= BS\,(2D^2 + 2D\,H_{kv}\,D_h) && (1) \\
M_{\text{attn}} &= BH\,(S^2 D_h + S^2 D_h) && \text{F1, two products per head} \\
&= 2BHS^2 D_h \\
&= 2BS^2 D && \text{A1: } HD_h = D \\
M_{\text{MLP}} &= BS\,(D\,D_{ff} + D_{ff}\,D) && \text{F1, A4} \\
&= 2BS\,D\,D_{ff}
\end{aligned}$$

MACs. Only the projection term contains H_{kv}. At $B = 1$:

Variant	M_{proj}	M_{attn}	M_{MLP}	Total
MHA	1,073,741,824	1,073,741,824	1,610,612,736	3,758,096,384
GQA	805,306,368	1,073,741,824	1,610,612,736	3,489,660,928
MQA	671,088,640	1,073,741,824	1,610,612,736	3,355,443,200

The QK and AV work stays fixed because H, S and D_h stay fixed. Every query head still forms its own scores and its own weighted sum of values, so the logical score tensor is still `[B,H,S,S]`, with the same softmax over the same entries. Fewer KV heads reduce the K and V projection work and the cache bytes. To report nominal FLOPs, with one fused multiply-add counted as two FLOPs, double each MAC total.

Speed and quality are measured

Real kernels may handle the grouping and the memory traffic differently, so end-to-end speed depends on the setting and the implementation. Any speedup has to be measured, and quality needs its own evidence.

Reported. Ainslie et al. found quality close to MHA for GQA in their setup of uptrained T5 models (Ainslie et al., 2023). At course scale you observe quality only as a training-loss comparison at matched steps and tokens, and Week 3 runs no task evaluation. The exercise's `kv_heads` comparison does this. It sets the loss curves and tokens per second of the `mha-rope`, `gqa-2` and `mqa` arms against their K and V parameters and cache bytes.

GQA in released models

GQA is a common choice in released decoders, but not a universal default. Llama 3 uses GQA with 8 KV heads at all three of its sizes, to improve inference speed and to reduce the size of the KV cache during decoding (Grattafiori et al., 2024). OLMo 3 uses MHA at 7B, with 32 query and 32 KV heads, and GQA at 32B, with 40 query and 8 KV heads (Ettinger et al., 2025). Week 4 keeps this divisible `n_kv_heads` configuration.

Causal Masking in Depth

One rule decides every allowed pair

$$\begin{aligned}\text{allowed}[b, q, k] = \text{valid_query}[b, q] \wedge (k \leq q) \\ \wedge \text{valid_key}[b, k] \wedge \text{same_document}[b, q, k]\end{aligned}$$

▼ A pair is allowed only when all four terms hold

Week 2 needed one mask, the causal one. Real batches add two complications: sequences of different lengths are padded to a common length, and several short documents are packed into one row. Both change which query-key pairs may interact, so the course writes one rule that covers causality, padding and packing.

Definition. The sequence index b selects a sequence in the batch, the query row q is the position whose output is being computed, and the key column k is a candidate position to read from. In this tab, q and k index rows and columns of the score matrix, not query and key vectors. Whether the pair (q, k) of sequence b is allowed is the conjunction of four Boolean terms:

$$\begin{aligned}\text{allowed}[b, q, k] = \text{valid_query}[b, q] \wedge (k \leq q) \\ \wedge \text{valid_key}[b, k] \wedge \text{same_document}[b, q, k].\end{aligned}$$

In words:

- the term $\text{valid_query}[b, q]$ holds when the query is a real token that may produce an attention output;
- the condition $k \leq q$ keeps the diagonal and blocks every future key, which is the causal mask of Week 2;
- the term $\text{valid_key}[b, k]$ holds when the key is a real source token;
- the term $\text{same_document}[b, q, k]$ holds when the query and the key belong to the same document.

The rule defines which pairs may interact. It says nothing about how a kernel computes them.

The two validity terms handle padding, and the document term handles packing. A conjunction does not depend on the order of its terms, so there is no precedence between separate masks to get right: there is only one mask.

SDPA takes one mask, and True means allowed Read on your own

Case	scaled dot-product attention (SDPA) call
plain causal	`attn_mask=None`, `is_causal=True`
padding or packing	one combined explicit mask, `is_causal=False`
additive mask	`0` allowed, `-inf` blocked

- In `nn.MultiheadAttention`, `True` means blocked (PyTorch Contributors, 2026c)
- Never pass a mask together with `is_causal=True` (PyTorch Contributors, 2026d)

▼ **Passing the mask to SDPA with the right polarity**

The exercise computes attention with PyTorch's scaled dot-product attention (SDPA). Build the mask once, broadcastable to the shape of the attention scores. For scores of shape `[B,H,S_q,S_k]` the mask is typically `[B,1,S_q,S_k]`, where B counts sequences, H query heads, S_q query positions and S_k key positions. Put the mask on the same device as the attention inputs, and build or cast an additive mask in a floating dtype the scores accept.

SDPA accepts the mask in two forms. A boolean mask is the rule itself: `True` means allowed and `False` means blocked (PyTorch Contributors, 2026d). An additive mask M is added to the scaled scores before the softmax, as in Week 2, with

$$M[b,q,k] = \begin{cases} 0 & \text{if allowed}[b,q,k], \\ -\infty & \text{otherwise.} \end{cases}$$

The module `nn.MultiheadAttention` uses the opposite polarity: in its boolean masks, `True` means blocked (PyTorch Contributors, 2026c). A boolean mask written for that module therefore cannot be passed to SDPA unchanged.

Plain causal attention may use `attn_mask=None` with `is_causal=True`. Any case with padding or packing passes the one combined explicit mask with `is_causal=False`.

Never pass an explicit mask together with `is_causal=True`. The SDPA documentation for the target version describes the two as mutually exclusive (PyTorch Contributors, 2026d). A check on 2026-07-14 found that the CPU math path accepted both arguments and composed them, and the CUDA backend the course selects has not been checked. The course rule, one explicit mask whenever there is one, holds whether a backend accepts the combination or rejects it.

Padding blocks keys, queries and targets Read on your own

- Block padded key columns
- Zero padded-query outputs after attention
- Skip padded targets in the loss

ILLUSTRATIVE PADDING



The causal mask alone is lower-triangular. The PAD query rows and PAD key columns are marked, but they keep their causal permissions.

With padding added, the PAD key columns and PAD query rows are blocked. The outputs of padded queries are explicitly zeroed after attention, and padded targets are excluded from the loss.

▼ Padding: block the keys, zero the outputs, skip the targets

Padding needs handling in three places: the key columns, the query rows and the targets.

Reading the figure

Both grids have one row per query and one column per key, over five real tokens t1 to t5 and two padding positions P. In the grid "causal only", $k \leq q$ alone lets the first padded query still read every earlier column. In the grid "causal + padding", the two validity terms mean that no real query reads padding and a padded query has no allowed key at all.

The output and the loss are separate steps

The mask decides only which values attention may combine, so the output and the target need two more steps after the attention call.

First, zero every padded query's attention output explicitly. With $O[b, q]$ the attention output of query row q, replace the padded rows with `torch.where` or `masked_fill`:

$$O[b, q] \leftarrow \begin{cases} O[b, q] & \text{if valid_query}[b, q], \\ 0 & \text{otherwise.} \end{cases}$$

Do not multiply the output by a 0/1 mask instead. A NaN the backend produced survives the multiplication, because $0 \cdot \mathrm{NaN} = \mathrm{NaN}$, while a replacement discards the old value.

Replacing the rows cleans the output but not the gradient. The gradient that reaches a replaced row is zero, but the backward pass of attention multiplies it by the derivatives of the NaN row, and $0 \cdot \mathrm{NaN} = \mathrm{NaN}$. The backend itself must therefore return finite rows.

Second, leave out of the shifted next-token loss every position whose target is padding. In this sequence that includes the step from the last real token t5 to the first P.

The output zeroing and the loss masking act on different tensors, so test each of them on its own.

A fully blocked row needs a policy Read on your own

- Test boolean and additive masks
- Fully blocked rows must come out finite and exactly zero
- Hand-written softmax: allow one sentinel key per blocked row

▼ Why a fully blocked row needs an explicit policy

Definition. Query row q of sequence b is **fully blocked** when $\mathrm{allowed}[b, q, k]$ is false for every key column k. A padded query row is fully blocked, because its false term $\mathrm{valid_query}[b, q]$ makes the conjunction false for every k.

The ordinary softmax divides zero by zero

Fix one sequence and one head. Write $u_{q,k}$ for the finite scaled score of query row q against key column k, and $a_{q,k}$ for its attention weight.

- **A1. An additive mask.** Every blocked pair gets $M[b, q, k] = -\infty$.
- **A2. The ordinary softmax.** Week 2's softmax over the key columns, with no special case for an empty row.

Recall from Week 2 that

$$e^{-\infty} = 0. \tag{F1}$$

Claim. Under A1 and A2, every weight of a fully blocked row is the undefined quotient $0/0$.

$$\begin{aligned} z_{q,k} &= u_{q,k} + M[b, q, k] = -\infty && \text{A1, every } k \text{ blocked} \\ e^{z_{q,k}} &= 0 && \text{F1} \\ \textstyle\sum_l e^{z_{q,l}} &= 0 && \text{a sum of zeros} \\ a_{q,k} &= \frac{e^{z_{q,k}}}{\sum_l e^{z_{q,l}}} = \frac{0}{0} && \text{A2} \end{aligned}$$

In floating point, this $0/0$ can produce NaN (PyTorch Contributors, 2026b). The weights of a row must sum to one over its allowed keys, and a fully blocked row has none, so the ordinary softmax has no valid answer for it.

Test the backend

Test fully blocked rows yourself, with a boolean and with an additive mask, on your device, dtype and selected backend. Require the outputs to be finite, and the padded rows to be

exactly zero after the output zeroing. Whether fully blocked rows come out finite depends on the device, dtype and backend, so you observe it in a test.

Finite stand-ins for $-\infty$

Some code writes a large finite negative number in place of $-\infty$, for example `torch.finfo(dtype).min`, the most negative value of the score dtype. For a row with at least one allowed key, the result is the same in practice: once the softmax subtracts the row maximum, the exponential of a blocked score underflows to 0. A fully blocked row then gives no $0/0$, because every score is finite. But its blocked scores are all about equal, so the softmax spreads its weight over keys the row may not read. A finite stand-in therefore replaces NaN by a finite but meaningless average, and the output zeroing and the loss masking are still needed.

Two cautions apply. The stand-in must be representable in the score dtype: -10^9, for example, lies far below the most negative FP16 value, about $-65{,}504$, and becomes $-\infty$ in FP16. And a stand-in added before the division by $\sqrt{d_k}$ shrinks with it, as Week 2's note on scaling before masking showed.

The sentinel key for a hand-written softmax

A hand-written softmax path that cannot represent an empty row safely needs one more rule: temporarily allow one sentinel key, the query's own position $k = q$. The denominator is then $e^{u_{q,q}}$, which is finite and positive, so the softmax puts weight 1 on that key and outputs the value vector at the padded position itself.

The sentinel is a numerical guard and grants no real permission. Zero that query's output explicitly after attention, so its arbitrary value cannot reach later computation, and leave its padded or boundary target out of the loss. This week's exercise has no hand-written softmax path, so it needs no sentinel.

The four safeguards are independent: the backend test, the sentinel, the output zeroing and the loss masking. Each guards against a different failure, and none of them replaces another.

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Which keys may B2 read?

Packed row: A1 A2 A3 B1 B2 B3

Answer A5

A sparse mask can still run dense Read on your own

- The mask sets allowed pairs
- Kernels may still compute every score
- Week 4: kernels that skip masked blocks (PyTorch Contributors, 2026a)

▼ **Allowed pairs and executed work are separate questions**

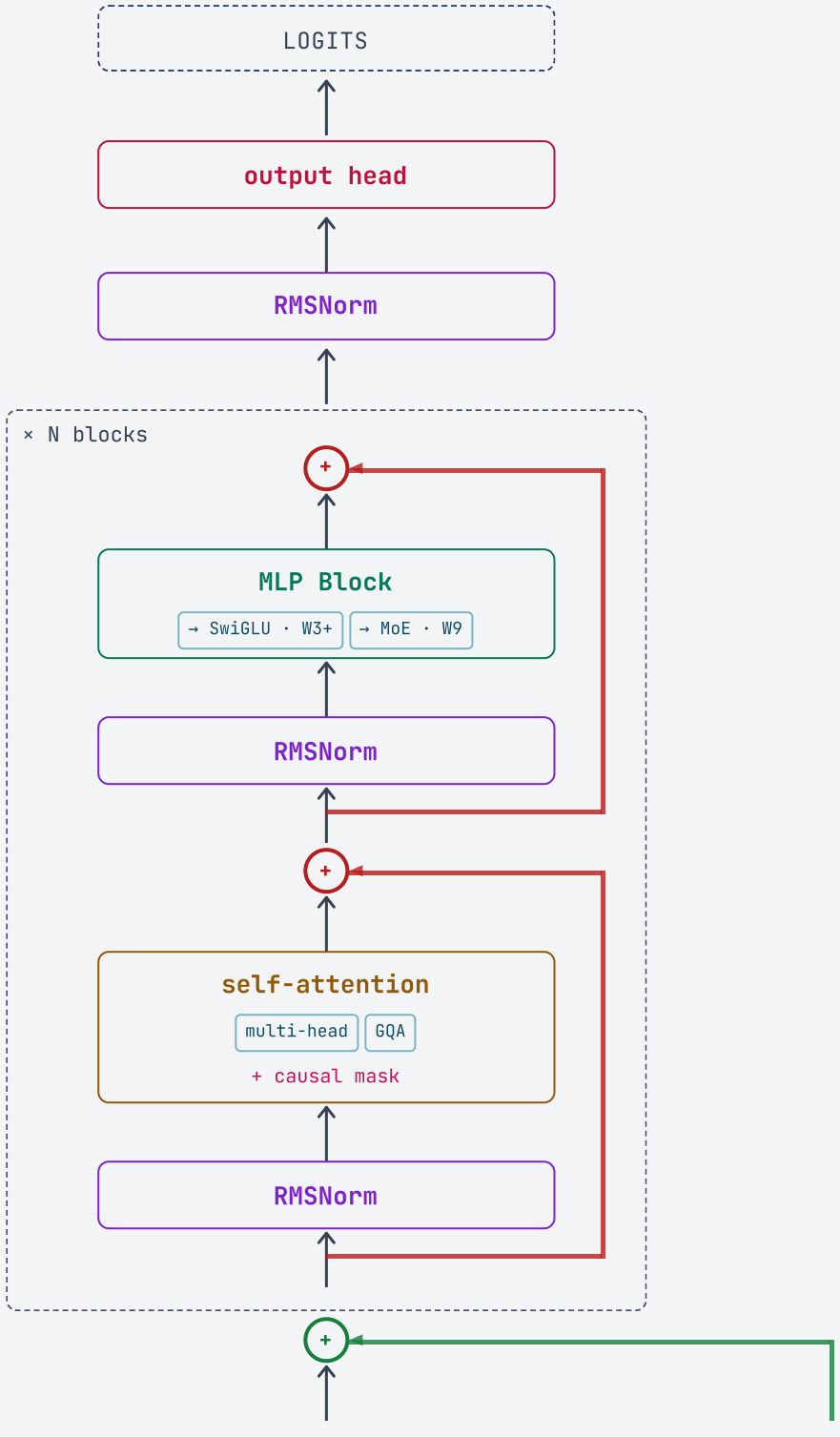
The allowed-pair pattern answers one question: which key values may influence each query. Executed work is a second question. With a dense explicit mask, a backend can compute every score, apply the mask or bias before the softmax, and then compute the full product of probabilities and values. The number of blocked cells therefore says nothing about speed.

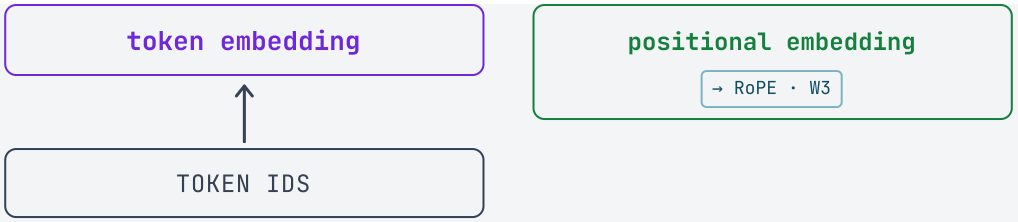
FlexAttention shows that block structure can be represented explicitly (PyTorch Contributors, 2026a). Week 4 covers how block representations map to skipped work, the I/O behavior of FlashAttention, and measured performance.

The Full Stack

Repeat the block, then normalize once

- The same block repeats `L` times, keeping `[B,S,D]`
- Final RMSNorm, then the tied head
- Later layers build on earlier ones





▼ From one block to a stack

One block performs one attention update and one MLP update. A language model needs a sequence of such transformations, so that representations can be refined step by step and circuits can compose across layers. Repeating Week 2's block changes what each block receives. Block 2 reads the residual stream that block 1 produced, block 3 reads the result of block 2, and so on. Attention in one layer can route information that a later MLP transforms, and a still later attention update can route the transformed features again. The induction pattern is one example of such composition. Depth makes it possible, but training decides which circuits appear, and no particular circuit is guaranteed.

One residual stream through the stack

Let B be the batch size, S the sequence length, D the model width and L the number of blocks, which the diagram labels `N`. Write x_ℓ for the residual stream of shape `[B,S,D]` that enters block ℓ, and a_ℓ for the stream after that block's attention update. **Definition.** Each block applies two pre-norm residual updates:

pre-norm block

$$a_\ell = x_\ell + \mathrm{Attention}(\mathrm{RMSNorm}(x_\ell))$$
$$x_{\ell+1} = a_\ell + \mathrm{MLP}(\mathrm{RMSNorm}(a_\ell))$$

Every term has shape `[B,S,D]`. The attention output projection returns to width D, and the MLP widens each position to D_{ff} and projects it back to D. Applying the two updates for $\ell = 0, \ldots, L-1$ carries one residual stream through all L blocks, from x_0 to x_L. Initialization, and how the residual stream grows with depth, are covered in Week 5.

The final norm and the tied head

Let V be the vocabulary size and E the `[V,D]` token-embedding matrix, which the tied output head reuses. After the last block, one final RMSNorm gives the stream z, still of shape `[B,S,D]`, and the tied projection gives logits of shape `[B,S,V]`:

$$z = \mathrm{RMSNorm}(x_L), \qquad \mathrm{logits} = zE^\top.$$

The final RMSNorm is outside the repeated block, so the stack applies it once, while each block has two RMSNorms of its own. A stack of L blocks therefore has $2L + 1$ RMSNorms.

The diagram shows this stack from the bottom up, and its arrows show which values depend on which, not a runtime schedule.

SwiGLU, the MLP variant the diagram points to

The diagram's MLP node carries a chip that points to SwiGLU, a gated variant of the MLP (Shazeer, 2020). The course MLP from Week 2 has two matrices, $D \to 3D \to D$, with GELU between them. SwiGLU adds a third matrix whose output gates the hidden vector, and Shazeer shrinks the hidden width to $\frac{2}{3}D_{ff}$ so that the parameter count and the computation stay those of the two-matrix MLP, as the Week 2 reading slide "Gated MLPs multiply two projections" derives.

Many recent models use SwiGLU. Llama 3 does (Grattafiori et al., 2024), and so do Qwen3 (Yang et al., 2025a) and OLMo 3 (Ettinger et al., 2025). DeepSeek-V3 uses it in its mixture-of-experts (MoE) layers (DeepSeek-AI, 2024), and gpt-oss in its experts (OpenAI, 2025). Gemma 3 uses the GELU-gated form, GeGLU (Google, 2025). The ungated two-matrix form remains in use as well. The GPT lineage uses it with GELU (OpenAI, 2019), and nanochat with a squared ReLU (Karpathy, 2026). The course keeps the two-matrix GELU MLP, and the exercise uses it unchanged.

The course recipe sets width and heads from depth Read on your own

depth recipe $D = 64L, \quad H = D/128, \quad D_{ff} = 3D$

L	D	H	D_ff	tied parameters
4	256	2	768	4.7M
8	512	4	1536	25.2M
12	768	6	2304	77.1M
16	1024	8	3072	176.2M

▼ Depth, width and the parameter count

Heuristic. In this example recipe, depth is the input: you choose the number of blocks L, and the recipe sets the model width D, the head count H and the MLP width D_{ff} from it. The rule for the width and the heads comes from nanochat (Karpathy, 2026), which computes

$$D = \left\lceil \frac{64L}{128} \right\rceil \cdot 128, \qquad H = \frac{D}{128}.$$

The course pairs the rule with its own MLP width, $D_{ff} = 3D$. The recipe is a heuristic for choosing example configurations, and it makes no claim about how loss scales. The course's model configuration, `ModelConfig`, does not compute this recipe.

Even and odd depths

Claim. For every even L, the rule gives $D = 64L$ and $H = L/2$. For odd L, it rounds $64L$ up to the next multiple of 128, which is $64(L+1)$.

The fraction inside the ceiling simplifies to $64L/128 = L/2$. For even L this is already an integer, so the ceiling leaves it unchanged:

$$\begin{aligned} D &= \left\lceil \tfrac{L}{2} \right\rceil \cdot 128 \\ &= \tfrac{L}{2} \cdot 128 \qquad L \text{ even, so } L/2 \text{ is an integer} \\ &= 64L, \\ H &= \tfrac{D}{128} = \tfrac{L}{2}. \end{aligned}$$

For odd L, the ceiling rounds $L/2$ up to $(L+1)/2$, and the rule gives $D = 64(L+1)$. Either way D is a multiple of 128, so H is an integer and every head has width $D_h = D/H = 128$. The slide's formula $D = 64L$ is the rule at even depths, and the table samples only even depths. Along the recipe, a deeper model is also a wider one.

The tied parameter count

The parameter column counts the course architecture at the recipe's widths. It is not nanochat's parameter count: the recipe takes only nanochat's rule for the width and the heads, and the blocks are the course's. Let P be the number of trainable parameters. The count rests on five properties of the course model.

- **A1. Tied embeddings.** One matrix E of shape `[V,D]`, with $V = 8192$, is both the token lookup and the output head.
- **A2. Attention.** Each block has bias-free multi-head attention with four square $D \times D$ projections W_Q, W_K, W_V and W_O.
- **A3. MLP.** Each block has a bias-free two-matrix GELU MLP, $D \to D_{ff} \to D$.
- **A4. Gains.** Each block has two RMSNorm gains of length D, and the stack has one more after its last block.
- **A5. Nothing else.** The model has no other trainable parameters.

Claim. Under A1 to A5, the tied parameter count is the exact identity

$$P = VD + L\left(4D^2 + 2D\,D_{ff}\right) + (2L+1)D. \tag{1}$$

Count each part on its own:

$$\begin{aligned}
P_{\text{emb}} &= VD && \text{A1: counted once, the head reuses } E \\
P_{\text{attn}} &= 4D^2 && \text{A2: per block} \\
P_{\text{MLP}} &= D\,D_{ff} + D_{ff}\,D && \text{A3: per block} \\
&= 2D\,D_{ff} \\
P_{\text{norm}} &= 2LD + D = (2L+1)D && \text{A4: all blocks and the final gain.}
\end{aligned}$$

By A5 these are all the parameters, so $P = P_{\text{emb}} + L\,(P_{\text{attn}} + P_{\text{MLP}}) + P_{\text{norm}}$, which is (1). It is Week 2's formula (9), there written with d_{ff} for the MLP width and evaluated for one block.

The four configurations

With $D_{ff} = 3D$, the two per-block terms add up to $4D^2 + 6D^2 = 10D^2$, and (1) becomes

$$P = VD + 10LD^2 + (2L+1)D.$$

At the recipe's four depths, with $V = 8192$, the three terms and their sums are:

L	D	VD	$10LD^2$	$(2L+1)D$	P
4	256	2,097,152	2,621,440	2,304	4,720,896
8	512	4,194,304	20,971,520	8,704	25,174,528
12	768	6,291,456	70,778,880	19,200	77,089,536
16	1,024	8,388,608	167,772,160	33,792	176,194,560

The slide's table rounds the totals to millions. The row $L = 8$ has the widths of the course configuration, $D = 512$ and $H = 4$, and its total is the 25,174,528 that Week 2 computed for eight blocks.

Depth alone

Moving from one row to the next changes L, D, H and D_{ff} together, so a comparison between rows cannot isolate the effect of depth. At fixed D, H and D_{ff}, by (1) each extra block adds $4D^2 + 2D\,D_{ff} + 2D$ parameters, while the tied embedding and the final gain are paid once.

Rotary Position Embeddings

The score should depend on the offset

- Score compares query `m` with key `n`
- Language structure mostly depends on the offset `n - m`
- Sinusoidal position is added once and mixed with content
- Nothing forces the score onto `n - m`

▼ **Position in the residual stream is a weaker guarantee than position in the score**

Attention scores a query at position m against a key at position n. Much of the structure of language depends on the offset $n - m$ between the two tokens, so a useful position signal should reach the score as that offset.

In Week 2, sinusoidal position entered the residual stream once, before the first block, as an elementwise addition to the token embedding. The sinusoidal basis also carries relative structure: Week 2 showed that shifting a position by a fixed offset rotates each frequency pair by an angle that depends on the offset and the pair but not on the position. The two facts are often read as a guarantee that the score sees the offset. Neither gives one.

To see why, write out the score of one head. Let $u_m, u_n \in \mathbb{R}^D$ be the normalized hidden states at positions m and n that enter the attention branch, and let $W_Q^{(h)}, W_K^{(h)} \in \mathbb{R}^{D \times D_h}$ be the query and key projections of head h. The head computes $q_m = W_Q^{(h)\top} u_m$ and $k_n = W_K^{(h)\top} u_n$, and its score before the $1/\sqrt{D_h}$ scale is

$$q_m^\top k_n = u_m^\top W_Q^{(h)} W_K^{(h)\top} u_n.$$

This is a bilinear form in the two hidden states, each of which mixes a token component and a position component in the same D channels. Nothing in the form makes the score depend on $n - m$ instead of on m and n separately. The model has to learn to recover the offset through W_Q and W_K, and whether it can depends on what they learn. Every layer that rewrites the residual stream can also weaken or overwrite the position component it inherited.

Make q·k depend only on n - m.

Rule: q may use only m, and k only n.

- Work in 2D: q and k as arrows
- What sets `q·k`: lengths, angle, both?
- What turns an arrow, keeps its length?

Rotating q and k leaves only the offset

column-vector rotation convention

$$R(\theta) = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$$

relative-rotation derivation

$$R(m\omega)^\top R(n\omega) = R(-m\omega)R(n\omega) = R((n-m)\omega)$$

relative-score identity

$$(R_m q)^\top (R_n k) = q^\top R_{n-m} k, \quad R_m = R(m\omega)$$

- Rotate Q and K, never V
- Full head: rotate all `D_h` coordinates, in pairs
- The identity holds for one score term (Su et al., 2024)

▼ Deriving the relative-score identity

The usual ideas

The task asks for a change to q that uses only the query position m and a change to k that uses only the key position n, such that their dot product depends on the positions only through $n - m$. Four ideas usually come up.

- **Add the offset to the score.** The rule forbids it, because q does not know n and k does not know m.
- **Scale by powers of a constant.** Multiply q by c^m and k by c^{-n} for some constant c. The product is $(c^m q)^\top (c^{-n} k) = c^{m-n}\, q^\top k$, so it depends on the positions only through the offset, and the idea is correct. But the lengths of the vectors, $c^m|q|$ and $c^{-n}|k|$, then grow or shrink with position, while a rotation keeps them.
- **Rotate.** Turn each vector by an angle proportional to its position.
- **Use a complex phase.** Read each two-dimensional vector as a complex number, and multiply q by $e^{im\omega}$ and k by $e^{in\omega}$. Multiplying by $e^{i\theta}$ turns a complex number by the angle θ and keeps its modulus, so this is the same rotation in other notation.

The rotation is the idea RoPE uses. It leaves the token content alone and rotates q and k after the head reshape. The offset then appears in the score by construction, instead of as something the projections must learn to expose. RoPE changes where position enters the computation, and it leaves the amount of position information as it was.

Assumptions and claim

Take one pair of coordinates of one head first.

- **A1. Fixed unrotated vectors.** The vectors $q, k \in \mathbb{R}^2$ are one coordinate pair of the query at position m and of the key at position n, after the head reshape. The derivation holds them fixed, so it concerns one score term.
- **A2. One frequency for both.** The query pair and the key pair turn at the same fixed angular frequency ω, in radians per position. RoPE takes the frequency from the pair index alone, so queries and keys share it.
- **A3. Angle proportional to position.** The query is rotated by $R_m = R(m\omega)$ and the key by $R_n = R(n\omega)$, where $R(\theta)$ is the counter-clockwise rotation on the slide, acting on column vectors.

Claim (exact identity). Under A1 to A3,

$$(R_m q)^\top (R_n k) = q^\top R_{n-m}\, k, \qquad R_{n-m} = R\big((n-m)\omega\big). \tag{1}$$

The positions enter the score only through the offset $n - m$.

The picture in two dimensions

Write $|q|$ and $|k|$ for the lengths of the two arrows, and φ_q and φ_k for their angles from the first axis, so that $q = |q|\,(\cos\varphi_q, \sin\varphi_q)^\top$ and $k = |k|\,(\cos\varphi_k, \sin\varphi_k)^\top$. The angle-difference identity of the cosine gives their dot product:

$$q^\top k = |q|\,|k|\,(\cos\varphi_q\cos\varphi_k + \sin\varphi_q\sin\varphi_k) = |q|\,|k|\cos(\varphi_k - \varphi_q).$$

The dot product depends on the two lengths and on the angle between the arrows. A rotation by θ keeps the length of a vector and adds θ to its angle. After both rotations the lengths are unchanged and the angles are $\varphi_q + m\omega$ and $\varphi_k + n\omega$, so their difference gains $n\omega - m\omega$:

$$(R_m q)^\top (R_n k) = |q|\,|k|\cos\big(\varphi_k - \varphi_q + (n-m)\omega\big).$$

The absolute angles $m\omega$ and $n\omega$ enter only through their difference.

The identity in matrices

The matrix version needs two facts about rotations. The first follows because the cosine is even and the sine is odd, the second from the angle-addition identities:

$$R(\theta)^\top = R(-\theta) \tag{F1}$$

and

$$R(a)\,R(b) = R(a+b). \tag{F2}$$

With these facts,

$$\begin{aligned}(R_m q)^\top (R_n k) &= q^\top R(m\omega)^\top R(n\omega)\, k && \text{transpose of a product, A3}\\ &= q^\top R(-m\omega)\, R(n\omega)\, k && \text{F1}\\ &= q^\top R\big((n-m)\omega\big)\, k && \text{F2}\\ &= q^\top R_{n-m}\, k && \text{definition of } R_{n-m}.\end{aligned}$$

The two absolute rotations meet in the product $R_m^\top R_n$, and that product depends on m and n only through $n - m$.

A worked pair

Take $q = (1, 2)^\top$, $k = (3, 4)^\top$, $m = 2$, $n = 5$ and $\omega = 0.1$, so that $m\omega = 0.2$, $n\omega = 0.5$ and $(n - m)\omega = 0.3$. Rotating each vector by its absolute angle and rotating the key once by the offset angle give the same score, up to the rounding of the coordinates:

$$\begin{gathered}(R(0.2)\, q)^\top (R(0.5)\, k) \approx (0.5827,\ 2.1588) \cdot (0.7150,\ 4.9486) \approx 11.0997,\\ q^\top R(0.3)\, k \approx (1,\ 2) \cdot (1.6839,\ 4.7079) \approx 11.0997.\end{gathered}$$

Multiplying out gives the exact value. For any angle θ, with the two coordinates of a pair numbered 1 and 2,

$$q^\top R(\theta)\, k = (q_1 k_1 + q_2 k_2)\cos\theta + (q_2 k_1 - q_1 k_2)\sin\theta,$$

which for this pair is $11\cos\theta + 2\sin\theta$. Both sides of (1) therefore equal

$$\begin{aligned}(R(0.2)\, q)^\top (R(0.5)\, k) &= q^\top R(0.3)\, k\\ &= 11\cos 0.3 + 2\sin 0.3 \approx 11.099741793704.\end{aligned}$$

Shifting both positions by the same amount leaves the score unchanged. At $m = 7$ and $n = 10$ the offset is still 3, and $(R(0.7)\, q)^\top (R(1.0)\, k) \approx 11.0997$ as well.

The full head is a sum over pairs

Week 3 rotates the full head: all D_h coordinates, in $D_h/2$ pairs, so D_h must be even. Write $q^{(j)} = (q_{2j}, q_{2j+1})^\top$ for pair j of the query, with $j = 0, \ldots, D_h/2 - 1$, and $k^{(j)}$ for pair j of the key. Pair j turns at its own fixed frequency ω_j, and R_m and R_n now stand for the rotation of the whole head, pair by pair. The dot product of two head vectors is the sum of the dot products of their pairs, and identity (1) applies to each pair with $\omega = \omega_j$:

$$
\begin{aligned}
&(R_m q)^\top (R_n k) \\
&\quad = \sum_{j=0}^{D_h/2-1} \big(R(m\omega_j)\, q^{(j)}\big)^\top \big(R(n\omega_j)\, k^{(j)}\big) && \text{each pair rotated on its own} \\
&\quad = \sum_{j=0}^{D_h/2-1} q^{(j)\top} R\big((n-m)\omega_j\big)\, k^{(j)} && (1) \text{ for each pair.}
\end{aligned}
$$

Every term of the sum depends on the positions only through $n - m$, and so does the whole rotary score of the head.

What the identity covers

Identity (1) is exact for one score term while the unrotated q and k are held fixed, as A1 states (Su et al., 2024). In the network, q and k are computed from the hidden states at positions m and n, and the residual states, the projections and the nonlinear layers can still carry absolute position into them. Each rotary score term is relative in this sense, but the network as a whole is not.

Only Q and K are rotated, never V. The values enter no dot product, so they pass unchanged into the weighted sum that the attention weights form.

The identity also covers the position reset of packed documents. If a document's positions are not reset, every position in it is shifted by the same constant Δ. A query at $m + \Delta$ and a key at $n + \Delta$ have the offset $(n + \Delta) - (m + \Delta) = n - m$. For full-head RoPE, the rotary score terms within the document depend only on that offset, so the shift leaves them unchanged.

Two ways to inject position Read on your own

- Sinusoidal: added once, at the input
- **RoPE**: rotates Q and K in every layer (Su et al., 2024)

comparing the two positional strategies:

sinusoidal PE (absolute)

- added ⊕ once at the input, before block 1
- encodes absolute position
- fixed, no learnable parameters
- influences later Q/K through hidden states

RoPE (inside attention)

- rotates Q and K inside the QK product, every layer
- makes each QK score term depend on offset (n - m)
- zero trainable parameters; cosine and sine values still computed or cached
- measured overhead depends on implementation

▼ Addition and rotation are different mechanisms

The two methods differ in where position first enters the computation, and how often.

With sinusoidal encoding, the position vector is added elementwise to the token embedding once, before the first block. At position m, with token t_m, the stream entering the first block is

$$x_0 = E_{t_m} + \mathrm{PE}_m \in \mathbb{R}^D,$$

where E_{t_m} is row t_m of the embedding matrix E and PE_m is row m of Week 2's fixed sinusoidal table. Later query and key projections receive positional information only through the hidden states that come from this input.

With RoPE, every attention layer first projects Q and K and reshapes them into heads. It then rotates their coordinate pairs at each token's position before the QK dot product (Su et al., 2024):

$$q_m \mapsto R_m\, q_m, \qquad k_n \mapsto R_n\, k_n \qquad \text{in every layer.}$$

Position therefore reaches every score directly, and each QK score term depends on the offset $n - m$. Under `pos_encoding: rope`, the course builds this rotation into every block.

Each method keeps a width unchanged, at a different place. Sinusoidal addition keeps the model width D, because E_{t_m}, PE_m and their sum all lie in $\mathbb{R}^D$. A rotation maps each head vector in $\mathbb{R}^{D_h}$ to another vector in $\mathbb{R}^{D_h}$, so RoPE keeps the head width D_h.

Each pair turns at its own frequency

pair frequency

$$\omega_j = b^{-2j/d_r}$$

pair rotation

$$R(m\omega_j) = \begin{pmatrix} \cos(m\omega_j) & -\sin(m\omega_j) \\ \sin(m\omega_j) & \cos(m\omega_j) \end{pmatrix}$$

- Base `b = 10000` in the course
- Pair 0 fastest, the last pair slowest
- Fixed angles, no learned parameters

▼ Full-head pairs and their frequencies

Which coordinates form a pair

Let $b > 0$ be the fixed RoPE base and d_r the even number of rotated coordinates in one head. Week 3 uses full-head RoPE, so $d_r = D_h$. The coordinates form $d_r/2$ interleaved pairs, $(0, 1)$, $(2, 3)$ and so on up to $(d_r - 2, d_r - 1)$, indexed by $j = 0, \ldots, d_r/2 - 1$. These are the adjacent columns $(2j, 2j + 1)$ of the exercise.

Some reference code pairs the columns differently. The `rotate_half` helper in Hugging Face Transformers pairs column j with column $j + d_r/2$ (Hugging Face, 2026). Both are rotary embeddings, but weights trained under one convention give wrong scores under the other, and the exercise's tests expect adjacent pairs. Bonus 2, the required reading on long-context extension, uses the same index j.

The base sets the frequencies

Definition. The first formula on the slide defines the angular frequency of pair j, in radians per position:

$$\omega_j = b^{-2j/d_r}, \qquad j = 0, \ldots, d_r/2 - 1.$$

The base and the frequencies are different quantities. The base b is one constant, and it sets how fast the frequencies fall with j. Each step from pair j to pair $j + 1$ multiplies the frequency by the same factor,

$$\frac{\omega_{j+1}}{\omega_j} = b^{-2/d_r},$$

which is less than 1 for $b > 1$. Pair 0 therefore turns fastest, at $\omega_0 = b^0 = 1$ radian per position, and the last pair, $j = d_r/2 - 1$, turns slowest, at $b^{-(d_r-2)/d_r}$. The course uses $b = 10000$, the default of `rope_theta` in its configuration.

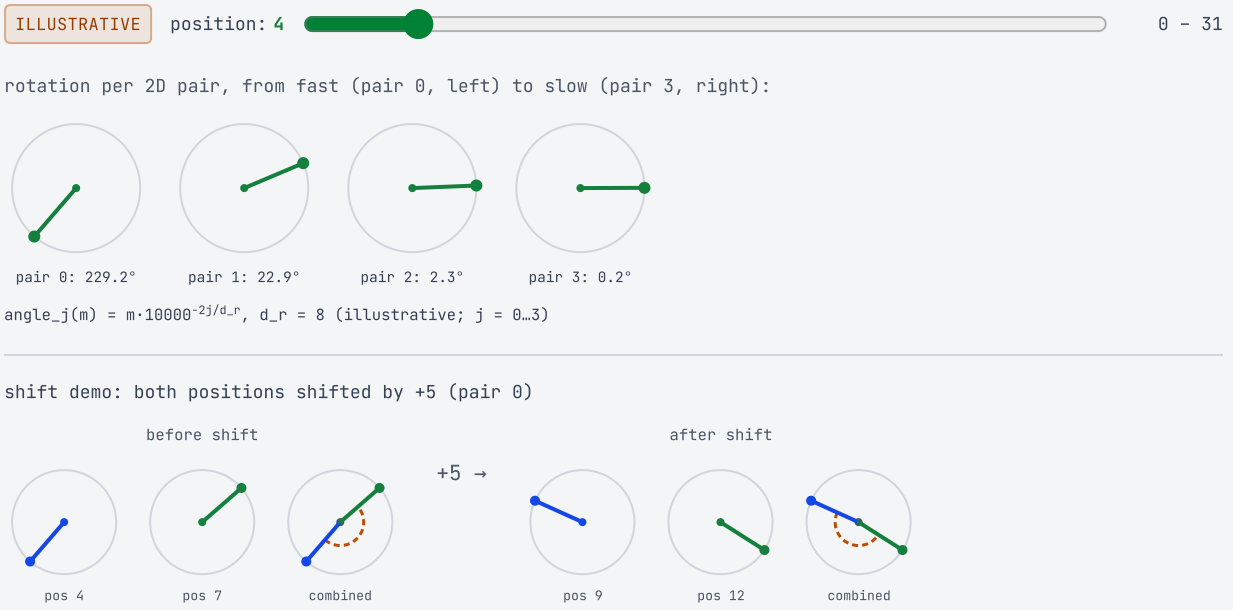
For $d_r = 8$ and $b = 10000 = 10^4$, the four frequencies are exact powers of ten. The last column gives each pair's wavelength $2\pi/\omega_j$, the number of positions it takes to turn once:

Pair j	Columns	Exponent $-2j/d_r$	ω_j (radians per position)	Wavelength $2\pi/\omega_j$ (positions)
0	$(0, 1)$	0	$10000^0 = 1$	6.28
1	$(2, 3)$	$-1/4$	$10000^{-1/4} = 0.1$	62.8
2	$(4, 5)$	$-1/2$	$10000^{-1/2} = 0.01$	628
3	$(6, 7)$	$-3/4$	$10000^{-3/4} = 0.001$	6,283

Nothing in these frequencies or rotation matrices is learned, so RoPE adds no trainable parameters. In the attention path, $R(m\omega_j)$ turns pair j of every query head and every key head at token position m, after the head reshape and before the scores are computed.

Fast pairs turn quickly, slow pairs barely move

- Pair 0 turns one radian per position
- Shifting both positions keeps the angle between the two vectors



▼ What the rotation dials show

The four dials show pairs $j = 0$ to 3 of an illustrative head with $d_r = 8$ and $b = 10000$, so their frequencies are 1, 0.1, 0.01 and 0.001 radians per position. The slider sets one position m from 0 to 31 for all four dials, and dial j shows the angle $m\omega_j$. Two positions show how far apart the bands are:

Pair j	ω_j	Angle at $m = 4$	Angle at $m = 31$
0	1	4 rad $\approx 229.2°$	31 rad $\approx 1776.2°$
1	0.1	0.4 rad $\approx 22.9°$	3.1 rad $\approx 177.6°$
2	0.01	0.04 rad $\approx 2.3°$	0.31 rad $\approx 17.8°$
3	0.001	0.004 rad $\approx 0.2°$	0.031 rad $\approx 1.8°$

At position 31, pair 0 has made almost five full turns, since $31/(2\pi) \approx 4.93$, and pair 3 has turned less than 2°.

The shift example below the dials uses pair 0 and the two positions $m = 4$ and $n = 7$. Shifting both by 5 moves them to 9 and 12. The offset stays $n - m = 3$, so the angle between the two rotated vectors stays $(n - m)\,\omega_0 = 3$ radians, about 171.9°, before and after the shift. The dials and vectors are synthetic coordinates chosen to show the rotation. They are neither learned Q and K values nor attention weights.

ILLUSTRATIVE position: 31 0 - 31

rotation per 2D pair, from fast (pair 0, left) to slow (pair 3, right):



pair 0: 1776.2° pair 1: 177.6° pair 2: 17.8° pair 3: 1.8°

angle_j(m) = m·10000$^{-2j/d_r}$, d_r = 8 (illustrative; j = 0…3)

shift demo: both positions shifted by +5 (pair 0)



enclosed angle (arc): 171.9° in both cases. Angle = (n - m) × base$^{-2\cdot0/8}$ = (7 - 4) × 1 = 3 rad, which depends only on the offset.

RoPE rotates each 2D pair by an angle proportional to the position. Pairs with a low index j rotate fast, and pairs with a high index rotate slowly. If you shift both positions by the same amount, the angle between the two vectors stays the same. For a fixed query and key, the score term of the pair shown here is therefore unchanged.

At position 31, pair 0 has made almost five full turns, while pair 3 has turned less than 2°.

RoPE adds no parameters but costs work Read on your own

- Parameters: zero
- Work: `O(BS(H+H_kv)d_r)` rotations
- Cosine and sine values are computed or cached
- 1–3% overhead in one fused setup (EleutherAI, 2021)

▼ Zero parameters still require rotary work

RoPE adds no trainable parameters, because its angles $m\omega_j$ are fixed. It still costs work, in every layer.

Counting the rotations

- **A1. What is rotated.** One layer rotates BSH query vectors and BSH_{kv} key vectors, in tensors of shape `[B,H,S,D_h]` and `[B,H_kv,S,D_h]`, and no value vectors.
- **A2. Rotated width.** Each vector has d_r rotated coordinates in $d_r/2$ pairs, with $d_r = D_h$ in Week 3.
- **A3. Cost of one pair.** Rotating one pair whose cosine and sine are known takes four multiplications and two additions, counting the 2×2 product as written, which a fused or `rotate_half`-style kernel arranges differently with the same six operations.

Claim (derived count). Under A1 to A3, the rotations of one layer take $3BS(H + H_{kv})\, d_r$ elementwise arithmetic operations, which is $O(BS(H + H_{kv})\, d_r)$:

$$\begin{aligned}\text{operations} &= \underbrace{BS(H + H_{kv})}_{\text{vectors}} \cdot \underbrace{d_r/2}_{\text{pairs per vector}} \cdot \underbrace{6}_{\text{per pair}} \qquad \text{A1, A2, A3} \\ &= 3\, BS(H + H_{kv})\, d_r.\end{aligned}$$

The two head counts enter separately, so under GQA, where H_{kv} is smaller than H, fewer key heads are rotated.

The cosine and sine tables

The sine and cosine values can be computed when needed or read from a table or a cache. The angle $m\omega_j$ depends only on the position m and the pair j, so sine and cosine arrays of shape `[S, d_r/2]` are enough. They broadcast across the batch and head axes, so they need not be stored once per token and head. Their memory and traffic depend on the table length, the cache policy, the dtype and kernel fusion. A table covers only the positions it was built for, so positions must stay within the configured range. Whether a model works at positions beyond its training length is a question for evaluation, which Bonus 2 takes up.

Measured cost and quality

Reported. EleutherAI reported roughly 1–3% overhead for its fused TorchScript rotary implementation, across the transformer sizes it tested (EleutherAI, 2021). RoPE's measured cost depends on kernel fusion, dtype, hardware, tensor layout and compilation, so other implementations, model sizes and hardware need their own measurement. The quality of RoPE and how well it extrapolates to longer contexts have to be measured as well (Su et al., 2024).

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Bonus 2 extends RoPE to longer contexts

Read on your own

- Position Interpolation: rescale positions (Chen et al., 2023)
- YaRN: a different remapping and adaptation (Peng et al., 2024)
- Bonus 2: read before the Week 4 tutorial

▼ **Long contexts change the position mapping**

Two methods extend a RoPE model to contexts longer than it was trained on. Position Interpolation rescales positions (Chen et al., 2023). YaRN uses a different remapping together with its own adaptation design (Peng et al., 2024). Both change how a position m maps to the rotation angles $m\omega_j$ of its pairs. Changing that numerical mapping does not by itself teach the model to use evidence from distant tokens. Bonus 2 covers both methods. It is required reading, and the Week 4 tutorial discusses its reflection questions, so read it before then.

NoPE & the Design Space

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No position signal at all.

Can a causal decoder tell where a token is?

Answer A6

31

Probes find position in NoPE models

- Probes recover position in the models studied (Haviv et al., 2022)
- **NoPE** generalizes best to longer sequences: ten tasks, 100M-parameter models (Kazemnejad et al., 2023)
- Your run: one training-loss comparison

▼ **What the NoPE evidence covers**

Haviv et al. ask whether causal language models trained without an explicit positional encoding still learn where each token is. **Reported.** Probes trained on the models' representations recover absolute position in the models they study (Haviv et al., 2022). The authors offer the growing prefix, which lets a model count predecessors, as a possible explanation. The study shows that position is recoverable in those models. It identifies no circuit that every causal decoder uses, and it does not show recovery at arbitrary lengths the models have not seen.

Kazemnejad et al. ask which positional scheme generalizes best from shorter training sequences to longer evaluation sequences. **Reported.** On ten controlled algorithmic and reasoning tasks, with decoder-only models of 100M parameters, NoPE does best in their comparison (Kazemnejad et al., 2023). The result holds for those tasks, that scale and that training. It does not show that NoPE matches the other schemes in frontier models at the training length.

Your own run adds a third kind of evidence. The exercise trains a `nope` arm beside the RoPE base and measures training loss, at one sequence length, one step budget and one model scale. Report those three with the result, and keep the claim to the training-loss comparison you measured.

Hybrids mix RoPE and NoPE across layers Read on your own

- Layer-wise hybrids: method evidence from one study (Yang et al., 2025b)
- MAI-Thinking-1: five local RoPE layers, then one global NoPE (Microsoft AI, 2026)
- One model's configuration

▼ Method evidence and one released configuration

Yang et al. study where in a model to use RoPE and where NoPE, layer by layer. **Reported.** In their evaluations, their hybrid models improve on long contexts and stay competitive on short ones (Yang et al., 2025b). This is evidence about the method for those models, under that training and those evaluations, and it supports studying layer-wise mixtures without giving a general recipe.

MAI-Thinking-1 is a dated example of adoption. **Reported.** Its official 2026 report documents a repeating pattern of five local RoPE layers with a 512-token window, followed by one global NoPE layer (Microsoft AI, 2026). A hybrid like this applies RoPE in selected layers only, while the course's `rope` setting rotates Q and K in every layer. The pattern also makes two kinds of choice at once: the window is a locality choice, and RoPE or NoPE is a position choice. MAI-Thinking-1 shows that one released model combines them in this way. It is one model's configuration and does not make the pattern a default. When such hybrids became common, how common they are, and whether this pattern beats others would take a broader survey to establish.

Position is a configuration choice

- `rope` : the base from Week 3 on
- `sinusoidal` : the Week 2 baseline
- `nope` : ablation and design option

Position and locality are separate choices

▼ Turning the choice into a controlled comparison

The setting `pos_encoding` selects one of three code paths:

`pos_encoding`	At the input	In every attention layer	Role in the course
`sinusoidal`	adds a position vector to the token embedding	nothing	the Week 2 baseline
`rope`	nothing	rotates Q and K after the head reshape	the base from Week 3 on
`nope`	nothing	nothing	an ablation and a design option

Only the RoPE path needs settings of its own, the rotary width and the frequencies, and the NoPE path puts no new positional operator in place of the two it removes.

Treat the three values as configurations of one controlled experiment, in which only the way position enters the computation varies. Hold everything else fixed: the architecture and parameter budget, the data order, the sequence length, the token budget, the optimizer and learning-rate schedule, the causal or packing mask, and the locality pattern. Record the mode, and the policy for resetting positions in packed documents, with the run, so that the comparison can be reproduced. Then compare the training-loss curves at the same number of steps and tokens.

Position handling (sinusoidal, RoPE or NoPE) decides how position reaches the attention scores. Sinusoidal encoding does it by changing the representations at the input. Locality changes which keys a query may attend to. They are separate design choices. A model can combine any choice of one kind with any choice of the other, so an effect of locality must not be credited to the positional mechanism. Week 4 varies locality.

Summary

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Six Takeaways

1. Heads: split, attend, concatenate, `W_O`; same `4D²`
2. Fewer KV heads: cache shrinks by `H/H_kv`
3. One composed mask; loss skips document boundaries
4. `L` pre-norm blocks, one final RMSNorm
5. RoPE: rotate Q, K; score sees offset
6. NoPE: mask gives some order; locality separate

▼ How attention reads several things at once, and knows where they are

The question for today was how attention reads several things at once, and how it knows where they are.

Several heads answer the first part. The projections split the D channels into H heads of width $D_h = D/H$, each head attends on its own, and the head contexts are transposed back, concatenated and mapped by W_O. Each query then gets H softmax rows for the $4D^2$ parameters of one wide head. Sharing key and value heads shrinks the KV cache by H/H_{kv} and leaves the attention work as it was. MHA, GQA and MQA form one family, set by a number H_{kv} that divides H. The cache saving is exact, and the effect on quality has to be measured.

One composed mask makes padded and packed batches correct, and the loss skips padded targets and those that cross a document boundary. The block repeats L times in pre-norm form, with one final RMSNorm after the stack.

For the second part, RoPE rotates the query and key coordinates of the full head in pairs, by angles proportional to position, so each rotary score term depends only on the offset $n - m$. It adds no parameters but costs work in every query and key head. A causal decoder without any position signal still gets some order from its mask, because the query at position t sees $t + 1$ keys. Locality, which keys a query may read at all, is a separate choice.

Six implementation checks

The six takeaways double as the exercise's checklist before you submit. Six implementation checks test the code behind them:

1. What shape must reach W_O? Concatenate the per-head contexts to restore `[B,S,D]`. The output projection W_O takes that model-width tensor.
2. Which rule aligns grouped KV heads with the query heads? Require that H_{kv} divides H, the check `n_heads % n_kv_heads == 0`. Each KV head then serves $g = H/H_{kv}$ consecutive query heads, and query head h reads KV head $\lfloor h/g \rfloor$.

3. How does a combined mask reach SDPA? As a boolean mask, in which `True` means allowed, and the composed mask goes with `is_causal=False`. An additive mask uses 0 for allowed and $-\infty$ for blocked entries.
4. Where is the last normalization? A final RMSNorm, after the loop over the blocks.
5. When is RoPE applied in the attention path? After the head reshape, rotate the paired Q and K coordinates before the scores are computed. The values are not rotated.
6. What does `pos_encoding: nope` select? No sinusoidal addition at the input and no rotary transform of Q and K. The causal mask and the other mask rules stay.

Work on the reflection questions without hints and bring your reasoning to the tutorial. Week 4 changes which keys are visible and how attention is executed: sliding windows, sinks, MLA, and FlashAttention.

To-dos until next week



Muddiest point

Do it right now! (2 min)

Before the tutorial

- Write down one question to discuss
- Think about the reflection questions

Self-study

- Read the reading slides and notes
- Read Bonus 2 before the Week 4 tutorial

Deadlines

- Quiz 1: Friday 30 October
- Second exercise: Wednesday 4 November, 22:00

Week 4 changes which keys are visible and how attention is executed: sliding windows, sinks, MLA, and FlashAttention.

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Reflection questions

1. At fixed model width, what changes when the number of attention heads increases?
2. Long-context serving is KV-cache bound, and a teammate proposes moving from four KV heads to one. Using this week's arithmetic, what exactly does that change and what does it leave unchanged? Name what the arithmetic cannot tell you, and say what you would run to bound it before shipping.
3. How should causal, padding, document-boundary, and loss masks compose?
4. A packed-document run trains for hours, then returns NaN loss on a batch whose last document is unusually short. Which two mask terms are your candidates, where would you place an assertion that separates them, and why did hours of healthy training not rule this out earlier?
5. Why does rotary position embedding encode relative offsets even though it rotates queries and keys by absolute positions?
6. You must choose `pos_encoding` for a long-context run this week. Your evidence is a controlled study over ten tasks with 100M-parameter decoders and one frontier model's reported layer configuration, and neither matches your workload. What does each source actually license you to conclude, which would you weight more heavily and why, and what would you measure on your own run to find out you were wrong?

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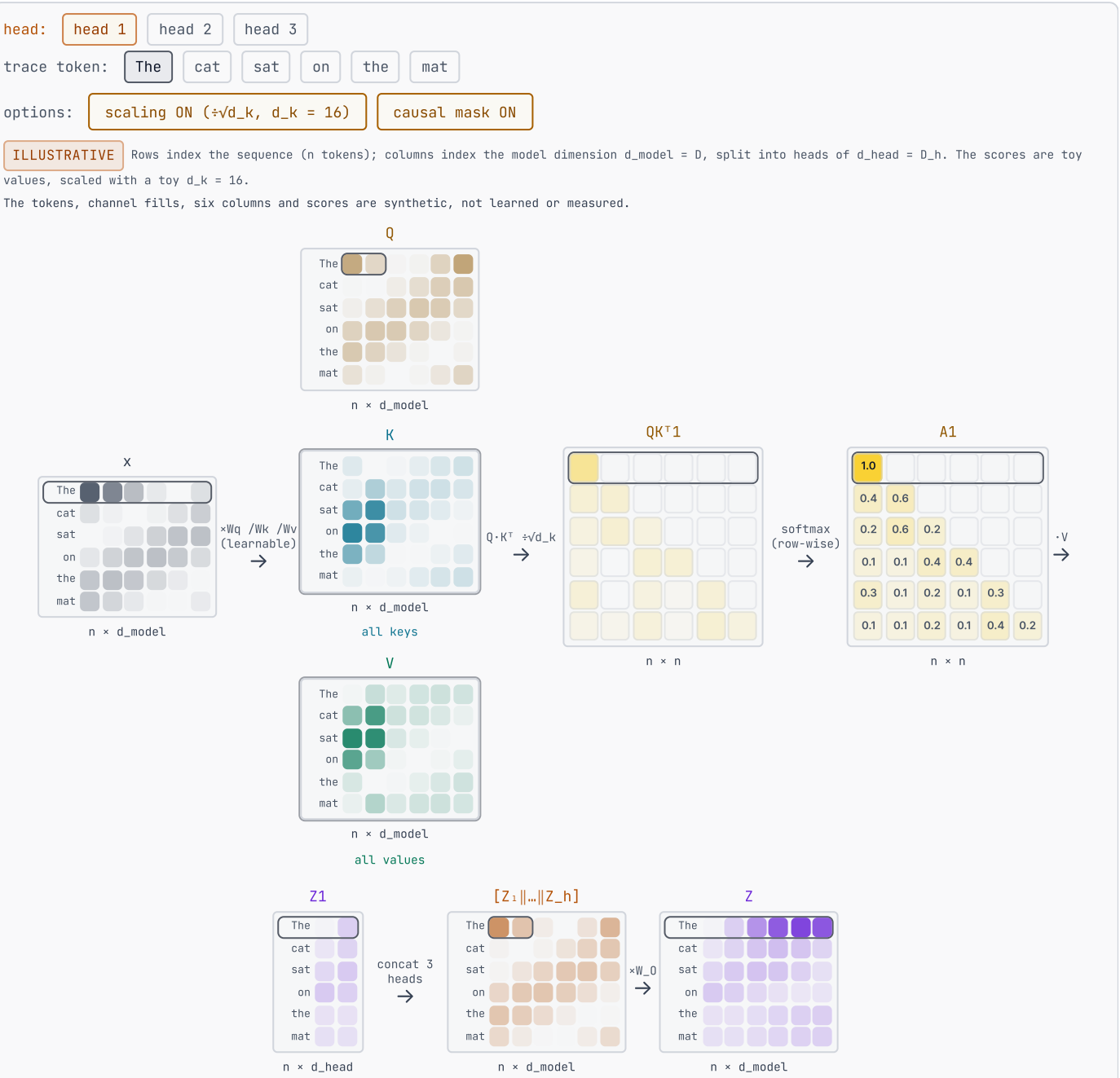
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- Ramelli, A. (1588). Water wheel driving pumps, one shown taken apart. *Le diverse et artificiose machine*, Figure LIII. Science History Institute. Public domain.
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Answers

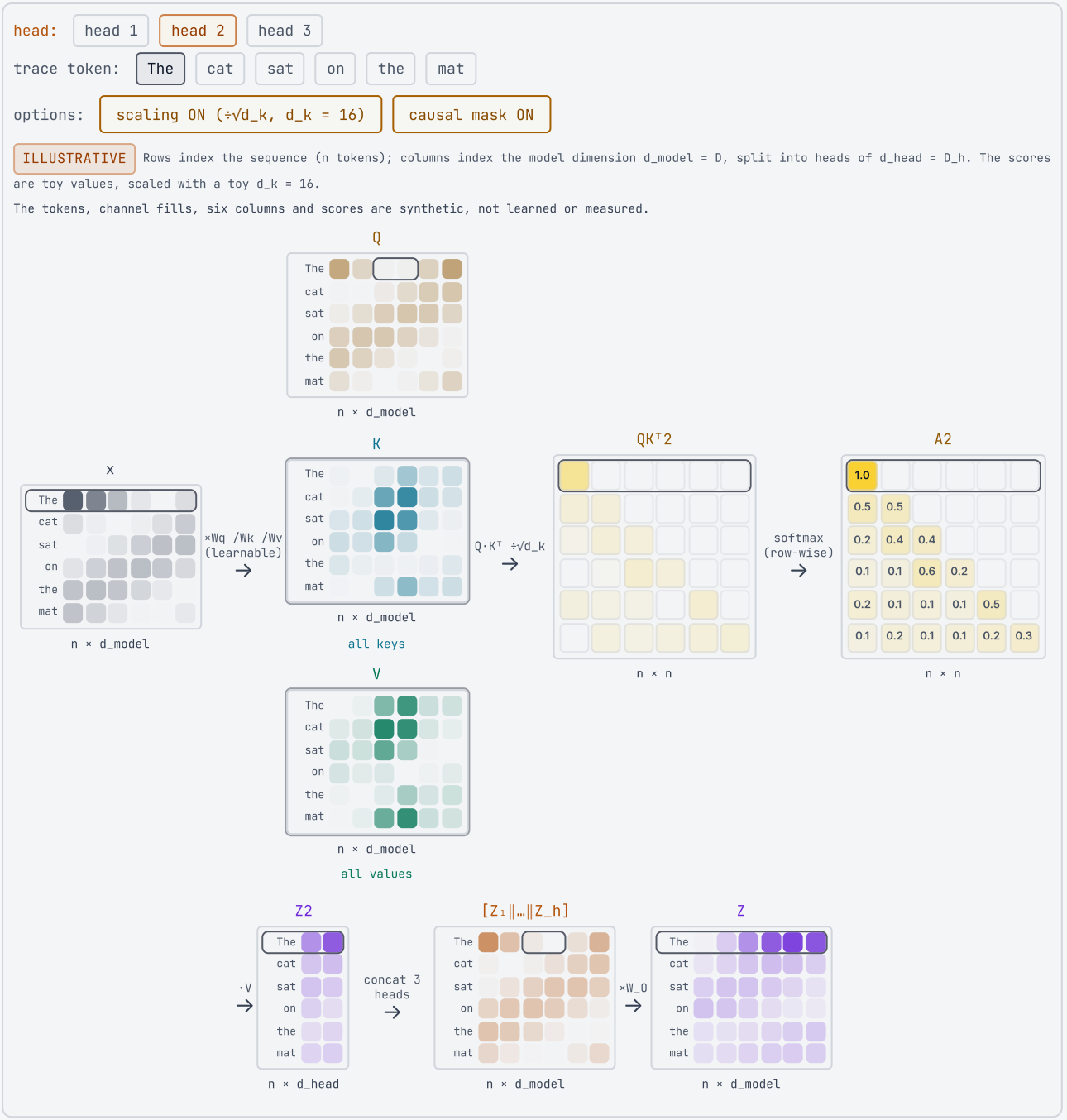
A1 · Several heads give several softmax rows

- Each head has its own projections and its own scores
- One softmax row per query, per head
- Concatenate the head outputs, then apply `W_O`
- No fixed job per head



Read left to right: each head computes softmax(QKᵀ ÷ √d_k) · V on its own channels. The input x is projected to Q, K and V. The scores QKᵀ go through a row-wise softmax, which gives the weights A. A then blends the values into the selected head's output. Concatenating the 3 head outputs and multiplying by W_O gives the final Z. The learned projections are D wide, and the 3 heads split them into D_h = D/H = 2 projected channels per head. In the code, the channels are reshaped from D into H × D_h and then transposed, so the head axis comes before the sequence axis. The diagram draws each head's query, key and value with D_h = 2 columns. Select a head first, then a trace token. The head control moves the highlight to the selected head's channels, now head 1's channels [0-1], and switches to that head's illustrative score pattern. These channels are a slice of the learned projections; each head sees all of x through its own weights. The token control selects the query row,

now the row of “The”. The query is compared with every key and blends every value, so all key and value rows stay available and K and V stay fully lit. QK^T, A and Z light up only the query’s own row. Toggle scaling to compare the scores before and after the division by √d_k. The scores are fixed toy values, not products of the drawn columns, so the scale uses a toy d_k = 16 whatever the drawn width. Scaling multiplies every score by 1/√d_k, so the QK^T panel dims when you turn it on. Toggle the causal mask to see the future keys (j > i) become unavailable. Change one control at a time, so that you can follow its effect along the whole path. The 3 head states are fixed illustrations, chosen to differ from one another. They do not show that real heads take on 3 fixed roles.



Head 2 on the same tokens: it reads its own slice of the channels, so its scores and attention rows differ from head 1's.

▼ Why one softmax row blends and several heads do not

A single attention head gives the query at position m one row of softmax weights a_{mn} over the keys n. Its output for that query is one weighted average of the value vectors,

$$\sum_n a_{mn}\, v_n, \qquad \text{with} \quad \sum_n a_{mn} = 1.$$

Suppose the query needs information from two positions n_1 and n_2. The head can split its weight between their two keys, say as a and $1 - a$, but its output is then

$$a\, v_{n_1} + (1 - a)\, v_{n_2},$$

one vector in which the two pieces of information arrive blended. One head can learn more than one behavior, yet for each query it produces one routing distribution.

Which ideas remove the blend

Several ideas come up when you try to remove the blend:

1. run attention several times, each time with its own query, key and value projections, so that each run computes its own scores and its own average;
2. split the model width into parts and attend within each part;
3. combine the separate results by concatenating or adding them;
4. stack a second attention layer;
5. make the softmax sharper, or keep only the two highest-scoring keys;
6. make the model wider.

The fifth and sixth ideas keep the blend. A sharper softmax or a top-two selection changes which keys receive weight, but the chosen keys are still averaged into one vector. A wider model gives that vector more channels, and each query still has one row. The fourth idea gives each query a second row, but one layer later and at the cost of depth. Within each layer, each query still has one row.

The first three ideas together are multi-head attention, with the split made in the projected channels. Each head has its own learned projections, so it computes its own score matrix, its own softmax row for each query and its own blend of the values. A query therefore gets one softmax row per head, and two heads can read two different positions without averaging them into each other. The outputs of the heads are concatenated, and an output projection W_O mixes them back into one vector for the residual stream.

No head has a fixed job. Neither the architecture nor the index of a head assigns one, and training decides what each head does.

What to compare in the attention widget

Keep the widget's trace token fixed and switch between its three heads: the same query gets its own softmax row in each head, because each head reads its own slice of the projected channels. With

scaling on, the widget divides fixed toy scores by $\sqrt{d_k}$ with $d_k = 16$, while each head of the model divides by $\sqrt{D_h}$, where the head width $D_h = D/H$ is the model width divided by the number of heads.

Back to the question

A2 · Four narrow heads cost the same as one wide head

Quantity	one width-`D` head	`H` width-`D_h` heads
projection parameters	$4D^2$	$4D^2$
dense attention work	$O(S^2D)$	$O(S^2D)$
channels per head	D	D/H

- Same parameters and work, but each head is narrower

▼ Counting parameters and work at fixed width

Neither layer has more parameters, and neither does more attention work. Both statements are exact counts under four assumptions, which hold everything fixed except how the width is split into heads.

- **A1. Fixed width.** The model width D is the same in both layers.
- **A2. The heads fill the width.** The heads together have $HD_h = D$ channels.
- **A3. Dense, bias-free projections.** Each of W_Q, W_K and W_V maps the D input channels to all HD_h head channels, and W_O maps them back to D.
- **A4. The same length, dense attention.** Every head forms its full $S \times S$ score matrix.

Parameters

Claim. Under A1 to A3, the four projections hold $4D^2$ parameters, whatever the number of heads H.

$$\begin{aligned} P_{\text{attn}} &= \underbrace{3 \cdot D \cdot HD_h}_{W_Q, W_K, W_V} + \underbrace{HD_h \cdot D}_{W_O} && \text{A3} \\ &= 3D^2 + D^2 && \text{A2} \\ &= 4D^2. \end{aligned}$$

The head count enters only through the product HD_h, which A2 fixes at D, so each of the four matrices is $D \times D$. At $D = 512$, both layers hold $4 \cdot 512^2 = 1{,}048{,}576$ attention-projection parameters.

Attention work

Count one multiply-accumulate as one MAC, which is two FLOPs. In one head and for one sequence, the scores $Q_h K_h^\top$ multiply an $S \times D_h$ matrix by a $D_h \times S$ matrix. Each of the S^2 scores is a dot product over D_h coordinates, so the product costs $S^2 D_h$ MACs. The blend multiplies the $S \times S$ weights by the $S \times D_h$ values V_h. It produces SD_h entries of S MACs each, again $S^2 D_h$. The work of one head is therefore proportional to $S^2 D_h$.

Claim. Under A2 and A4, the two attention products of one layer cost $2S^2D$ MACs per sequence, whatever the number of heads H.

$$
\begin{aligned}
M_{\text{attn}} &= H \cdot 2S^2 D_h && \text{A4: } H \text{ heads, two products each} \\
&= 2S^2 \cdot H D_h \\
&= 2S^2 D && \text{A2.}
\end{aligned}
$$

One head of width D is the case $H = 1$ and $D_h = D$, with the same $2S^2 D$, so both layers do $O(S^2 D)$ attention work, and a batch of B sequences multiplies the count by B. This counts matrix-multiply work and is not a model of latency. Kernels, memory traffic and how well the hardware is used can still make the two layers run at different speeds.

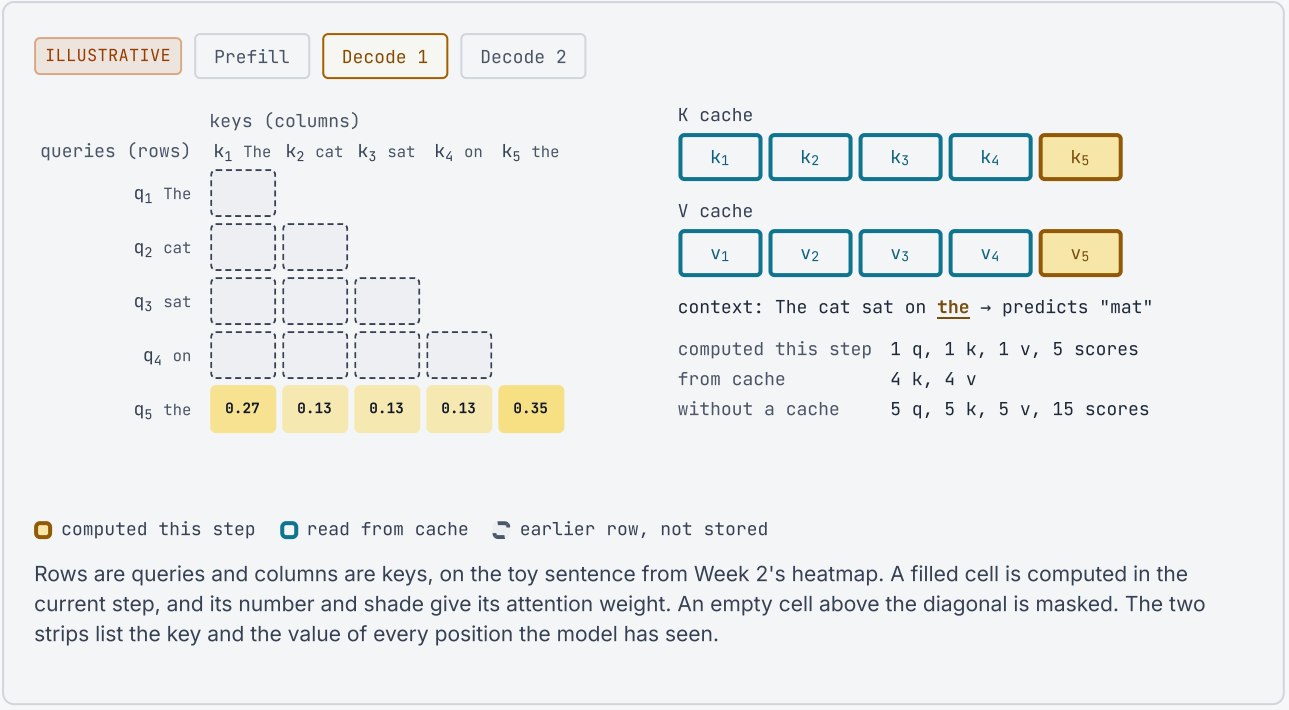
The same budget, split differently

The two layers spend the same budget in different ways. With H heads, each query gets H rows over $D_h = D/H$ channels each. Equal cost does not mean unlimited capacity or better quality: at a fixed width, every extra head makes each head narrower, and the counts say nothing about which split trains the better model.

Back to the question

A3 · Decode adds one row and reads the cache

- Prefill computes all rows of the prompt at once
- **Decode** adds one new query row per token
- **Key-value (KV) cache**: keys and values of past tokens
- Earlier rows are never stored and never recomputed



Decode step 1: the query of "the" forms one new row against all five keys. Four keys and values come from the cache, and one key and one value are new.

▼ Prefill computes the prompt, decode adds one row per token

A language model generates text autoregressively: it predicts one token, appends it to the sequence and predicts again from the longer sequence. Generation starts from a prompt and runs in two phases. Prefill handles the prompt, and decode produces the new tokens one at a time. The widget shows both on the sentence of Week 2's heatmap, "The cat sat on the mat", with "The cat sat on" as the prompt.

A full model repeats Week 2's block several times, as a stack of layers. The output of one layer at a position is the input of the next layer at the same position.

Prefill runs the prompt through the model once. In every layer it computes the query, key and value of every prompt position. Its causal score matrix is the same lower triangle as in the forward pass of training: each query row holds the scores of its own key and of the keys before it. The output of the last row predicts the first new token. Prefill also writes every key and value it computed into the **key-value (KV) cache**, which keeps them for each layer and each head. In the widget, prefill computes 4 queries, 4 keys, 4 values and the 10 scores of the 4×4 triangle, and the row of "on" predicts "the".

Decode then feeds one token per step, the token just predicted. In every layer, the step computes the query, key and value of this one new position. It appends the new key and value to the cache and scores the one query against every key in the cache, which adds one new row to the score matrix. The softmax weights of this row then blend all the values in the cache. Only the new position's output goes on. It forms the new position's query, key and value in the next layer, and after the last layer it predicts the next token.

The earlier rows are never needed again, because their scores and weights formed only their own outputs, and those outputs reach the new token only through the keys and values already in the next layer's cache. So the cache stores no scores, and no step recomputes an earlier row.

Why the cached keys and values are exact

The cache changes nothing in the result, and the causal mask is the reason. Number the positions from 1.

Claim. In every layer, the key and value of position i depend only on the tokens at positions 1 to i.

In the first layer, the key and value of position i are projections of its input vector, which depends only on token i and its position. Attention at position i reads only the keys and values of positions up to i, because the mask hides every later one. The other parts of the block, the normalization, the MLP and the residual addition, act on each position separately. So the output of position i depends only on the tokens at positions 1 to i, and so do the key and value it forms in the next layer. The same step, repeated layer by layer, gives the claim for every layer.

A token appended later therefore cannot change a cached key or value. The cached vectors equal those of a full recomputation of the longer sequence, so the new row has the same weights as the matching row of the full causal score matrix. Without the causal mask, the argument fails. From the second layer on, an earlier position would then read the appended token, so its key and value would change with every new token, and the cache would hold outdated vectors.

What the cache saves and what it costs

Decode step 1 in the widget feeds "the", the fifth token. The step computes 1 query, 1 key, 1 value and 5 scores, and it reads 4 keys and 4 values from the cache. Without a cache, each step would recompute the whole sequence: 5 queries, 5 keys, 5 values and the 15 scores of the 5×5 triangle. These counts are per head and per layer.

In exchange, the cache grows by one key and one value per token, per layer and per key-value head (in MHA, every head), for as long as the sequence runs. Week 10 covers serving: how a server schedules prefill and decode steps across requests and manages the cache's memory.

Reading the widget

In the decode states, compare the one computed row at the bottom with the faded earlier rows above it. In the two strips, compare the one filled key and value, computed in this step, with the outlined ones, read from the cache. Week 2's heatmap computes the full causal matrix of the same sentence, and its rows for "the" and "mat" show the same weights as the decode rows. The split into prefill and decode,

and the marking of each key and value as computed in the step or read from the cache, follow a figure by Verma and Vaidya (Verma & Vaidya, 2023).

Back to the question

A4 · Here the KV cache is ten times the weights

- KV cache: 512 MiB
- Weights: about 48 MiB
- Here the cache is about 10× larger

▼ Weights against cache in one serving setup

In this setup the key-value (KV) cache, which holds the keys and values of every past token, needs more memory than the weights: 512 MiB against about 48 MiB, roughly ten times as much (the exact ratio is $536{,}870{,}912/50{,}349{,}056 \approx 10.7$).

Definition. Take B sequences of S cached tokens and a model of L layers, each with H_{kv} key-value heads of width D_h. The **KV-cache payload** is the number of cached elements,

$$N_{\mathrm{KV}} = 2\,B\,S\,L\,H_{kv}\,D_h,$$

where the factor 2 counts keys and values. In bytes, the payload is N_{KV} times the bytes per element.

The course base model has $L = 8$ layers and $H_{kv} = 4$ key-value heads of width $D_h = 128$. The setup serves $B = 32$ sequences of $S = 1{,}024$ tokens in BF16, at 2 bytes per element, and memory is counted in MiB, with $1\,\mathrm{MiB} = 2^{20}$ bytes. The weights are the model's 25,174,528 parameters, which the Full Stack tab counts term by term, stored at 2 bytes each as well:

Stored tensors	Elements	Bytes	MiB
KV cache, $2 \cdot 32 \cdot 1024 \cdot 8 \cdot 4 \cdot 128$	268,435,456	536,870,912	512
Weights	25,174,528	50,349,056	≈ 48

The ratio belongs to this setup. The cache grows with the batch and the context, and the weights stay the same size. A large model serving one short sequence is dominated by its weights instead.

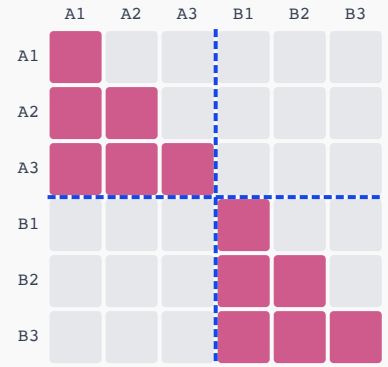
Back to the question

A5 · Packed documents never see each other

- B2 reads B1 and B2 only
- Block-diagonal: one block per document
- Restart positions per document, skip targets across boundaries
- Rebuild the mask for every batch

ILLUSTRATIVE PACKED SEQUENCES

packed sequence (doc A + doc B)



Each document attends only within itself. Position IDs restart at each document, and the next-token targets that cross a document boundary are masked.

allowed masked doc boundary

▼ Packed documents: attention, positions and the loss

B2 may read B1 and B2. Both are in its own document, and neither is in its future. The A tokens are in another document, and B3 comes after B2.

The block-diagonal mask

Applied to every query, the rule gives one lower-triangular block per document on the diagonal, and every cell outside those blocks is masked. A mask of this form is called **block-diagonal**. In the figure, a dashed line separates two illustrative three-token documents, A and B.

A worked row with padding

Take a six-slot row with packed IDs `[A,A,A,B,B,PAD]` and reset positions `[0,1,2,0,1,0]`. Under the four-term rule its slots allow the keys below, and the last two columns give the shifted next-token targets and whether the loss keeps them.

Slot	Token	Reset position	Allowed keys	Next-token target	In the loss
0	A1	0	A1	A2	yes
1	A2	1	A1, A2	A3	yes

Slot	Token	Reset position	Allowed keys	Next-token target	In the loss
2	A3	2	A1, A2, A3	B1	no: across the document boundary
3	B1	0	B1	B2	yes
4	B2	1	B1, B2	PAD	no: the target is padding
5	PAD	0	none	none in this row	no

Recall Week 2's loss over a set $\mathcal{T}$ of retained pairs (b, i), which averages over those pairs only:

$$\mathcal{L} = -\frac{1}{|\mathcal{T}|} \sum_{(b,i)\in\mathcal{T}} \log p_\theta(x_{b,i+1} \mid x_{b,\le i}).$$

For a packed row, $\mathcal{T}$ leaves out both the transition from A3 to B1, whose target lies across the document boundary, and the transition from B2 to PAD, whose target is padding. Tests therefore cover the attention rows, the reset positions and the loss transitions together.

Positions restart in each document

The position reset is the subtler part, because leaving it out crashes nothing. Positions restart in each document because a document's meaning does not depend on where it happened to land in the packed row. Without a reset, the first token of document B carries position 3 in this row, and position 0 in a batch that packs B first. An absolute position signal then sees the same text differently depending on its neighbors.

Whether a missing reset matters depends on the position mechanism. Sinusoidal encoding adds the position to the input vector, so a missing reset changes the input itself and every score downstream. Under full-head RoPE, the rotary terms of the scores inside a document depend only on offsets (the RoPE tab shows this), so a missing reset leaves them unchanged. NoPE uses no positions. Wherever rows are packed, every arm of a comparison must receive the same document-local positions, or the comparison is not controlled. In this week's exercise the tests check `document_position_ids` and the mask path, while the training runs use unpadded windows of one token stream. Every arm that uses positions therefore sees the default positions $0, \ldots, S-1$.

The mask changes with every batch

The document boundaries depend on what each batch packs. Carry per-batch document metadata into the mask construction, or rebuild the mask whenever the boundaries change. A fixed mask reused for a different packing layout can allow the wrong pairs without any error.

Back to the question

A6 · A causal mask alone carries some position

- Partly: query `t` sees `t + 1` keys
- Without the mask, attention is order-blind

ILLUSTRATIVE Pattern: **RoPE** NoPE

SYNTHETIC ROPE MATRIX: IDENTICAL CONTENT, COSINE SCORES

	The	cat	sat	on	the	mat
The	1.00					
cat	0.42	0.58				
sat	0.18	0.35	0.48			
on	0.10	0.16	0.31	0.43		
the	0.12	0.09	0.14	0.27	0.38	
mat	0.19	0.10	0.08	0.11	0.22	0.31

Both matrices are synthetic and give every token the same content, so only the position handling differs. Under NoPE every logit is the same, and each row spreads its weight uniformly over its causal prefix. Under RoPE each logit before the softmax is the cosine of the relative offset, so the weights rise and fall with the phase of the rotation. RoPE does not build in an ordering by recency.

ILLUSTRATIVE Pattern: RoPE NoPE

SYNTHETIC NOPE MATRIX: IDENTICAL CONTENT, UNIFORM SCORES

	The	cat	sat	on	the	mat
The	1.00					
cat	0.50	0.50				
sat	0.33	0.33	0.33			
on	0.25	0.25	0.25	0.25		
the	0.20	0.20	0.20	0.20	0.20	
mat	0.17	0.17	0.17	0.17	0.17	0.17

Both matrices are synthetic and give every token the same content, so only the position handling differs. Under NoPE every logit is the same, and each row spreads its weight uniformly over its causal prefix. Under RoPE each logit before the softmax is the cosine of the relative offset, so the weights rise and fall with the phase of the rotation. RoPE does not build in an ordering by recency.

The NoPE setting: every logit is equal, so row 0 puts all its weight on its own key and row 5 puts 1/6 on each of its six keys.

▼ What causality changes, and what it does not prove

Partly. NoPE, for no positional encoding, names a causal decoder with no position signal at all: no position vector at the input and no rotation of queries and keys. Two facts decide what such a model can know about order. Without its mask, self-attention is blind to order. With the causal mask, different query rows see prefixes of different sizes.

Without the mask, attention is order-blind

Week 2 proved order-blindness for one head, on the slide "Without position, attention sees a set". Two of its facts carry over. Let P be an $S \times S$ permutation matrix, X the input states of one sequence, and $A(\cdot)$ the row-wise softmax. For every score matrix Z,

$$A(PZP^\top) = P\,A(Z)\,P^\top, \tag{F1}$$

and for one head without a mask or a position signal,

$$\mathrm{Attention}(PX) = P\ \mathrm{Attention}(X). \tag{F2}$$

Claim. Without a mask and without a position signal, multi-head attention is permutation-equivariant along the sequence axis: permute the input tokens, and the outputs are permuted in the same way.

Each head h computes its context C_h from its own slices of the projections, so F2 applies to every head. The heads are then concatenated along the width and multiplied by W_O from the right:

$$
\begin{aligned}
&C_h(PX) = P\,C_h(X) && \text{F2, for each head } h \\
&C(PX) \\
&\quad = \left[\, P\,C_0(X), \ldots, P\,C_{H-1}(X) \,\right] && \text{concatenation along the width} \\
&\quad = P\,C(X) \\
&C(PX)\,W_O = P\,\big(C(X)\,W_O\big) && \text{associativity.}
\end{aligned}
$$

Permuting the tokens only reorders the output rows, so the layer itself has no notion of order.

The causal mask breaks the symmetry

With the causal mask, a head's scores are $Z + M$, where M is Week 2's additive mask: $M_{tn} = 0$ for $n \le t$ and $M_{tn} = -\infty$ for $n > t$. The mask is attached to the positions, so it stays in place when the tokens are permuted, and the permuted input gives the scores $PZP^\top + M$. Week 2's step "softmax under permutation" uses F1, which needs the scores $P(Z + M)P^\top = PZP^\top + PMP^\top$, so it applies only if $PMP^\top = M$. That matrix is the causal mask with its rows and columns reordered, and it equals M only for the reordering that keeps every pair of positions in order, the identity permutation, so for every other permutation the proof fails. Two tokens show that the property itself fails, not only the proof. In the order (a, b), the first output is v_a, because the first query sees only itself. Swap the tokens, and the first output is v_b. Equivariance would require it to equal the second output of the original order, a blend of v_a and v_b, which differs from v_b whenever $v_a \neq v_b$.

What the mask adds is a different prefix for every row. The query at position t may read its t predecessors and itself, the keys at positions $0, \ldots, t$, so it sees $t + 1$ keys.

Uniform weights over the prefix

The widget has two settings, RoPE and NoPE. Its NoPE setting is the case of the question, and three assumptions describe it.

- **A1. No position signal.** A score depends on the content of its query and key alone.
- **A2. Identical content.** Every query is the same vector q, every key the same vector k and every value the same vector v.
- **A3. The causal mask.** The query at position t reads only the keys $n \le t$, and masked entries get the score $-\infty$.

Claim. Under A1 to A3, row t of the attention weights is uniform over its prefix:

$$a_{tn} = \begin{cases} \dfrac{1}{t+1} & n \le t, \\ 0 & n > t. \end{cases}$$

Write z_{tn} for the scaled, masked score of query t and key n. By A1 and A2, every visible score is the same number $c = q^\top k / \sqrt{D_h}$. For a visible key, $n \le t$,

$$\begin{aligned} a_{tn} &= \frac{e^{z_{tn}}}{\sum_{n'=0}^{t} e^{z_{tn'}}} && \text{softmax over the visible keys, A3} \\ &= \frac{e^{c}}{(t+1)\, e^{c}} && \text{A1 and A2: every visible score is } c \\ &= \frac{1}{t+1}, \end{aligned}$$

and a masked key, $n > t$, gets the weight $e^{-\infty} = 0$.

The constant c cancels, so the weights do not depend on the content at all. The rows differ only in the size of the prefix the mask gives them. In the widget's six rows, row 0 puts its whole weight on its own key, and row 5 puts $1/6$ on each of its six keys.

The outputs of the head, however, are all equal in this setting, because under A2 each is a weighted average of copies of v, which is v itself. The prefix size reaches a head's output only once the contents differ. If the first token is a distinct start token with value v_0 and every other token has value v, the query at position t outputs

$$\frac{1}{t+1}\, v_0 + \frac{t}{t+1}\, v,$$

which changes with t.

The RoPE setting

The RoPE setting uses one rotary pair, a head of width $D_h = 2$ with $\omega = 1$, and the same unit vector q as the query and the key at every position. Writing the query position as m, as the RoPE tab does, the query at m and the key at n have the scaled logit

$$z_{mn} = \frac{\cos\big((n-m)\omega\big)}{\sqrt{2}}.$$

This follows from the relative-score identity with $k = q$: the score becomes $q^\top R\big((n-m)\omega\big)\, q$, which is $\cos\big((n-m)\omega\big)$ because $|q| = 1$, and the scale $1/\sqrt{D_h} = 1/\sqrt{2}$ gives z_{mn}.

Both settings are controls, and neither predicts the attention of a trained model, whose queries and keys depend on the content, differ in length and direction, and combine many rotary frequencies. Its learned scores depend on the content as well as on position, and they can oscillate with the offset.

What the prefix signal can and cannot do

The different prefix sizes are a signal that training can exploit. A network could, for example, read it from the weight $1/(t+1)$ that a query puts on a distinct start token, which amounts to counting its t predecessors. Three questions remain empirical: whether a trained layer actually counts predecessors, whether the representation identifies every absolute position uniquely, and whether the signal stays reliable at lengths the model has not seen.

Back to the question